Exchange Rates and International Risk Sharing under Home Portfolio Bias[*]

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December 11, 2025

This is a preliminary draft.

Abstract

The standard view in international macroeconomics is that exchange rate dynamics are inconsistent with the notion that international financial markets enable countries to share risk effectively. I develop a model of international financial frictions that is consistent with two facts about international asset holdings: (i) home portfolio bias and (ii) the elasticity of substitution in international portfolio choice, and I show that these portfolio facts characterize the extent to which countries share risk in equilibrium. When matched to observed portfolio allocations and elasticities, the model implies extensive international risk sharing, yet when combined with observed volatility in international trade expenditure shares, it solves the key Backus-Smith exchange rate puzzle, which is that a country's consumption increases when its consumption bundle becomes more expensive (a real exchange rate appreciation). In the model, an increase in the international relative demand for a country's goods raises their price and increases their firms' profits; under home portfolio bias, it also raises the relative income of domestic households, who own most of the country's firms, so they consume more. More generally, this mechanism delivers the procyclical, volatile, and persistent exchange rates seen in the data, whereas other popular shocks in the literature cannot do so when matched to observed portfolios.

[*]I would like to thank my advisors Patrick Kehoe, Elena Pastorino, Luigi Bocola, as well as Adrien Auclert, Adrien Bilal, Sebastian Di Tella, Chris Tonetti, Arvind Krishnamurthy, Rafael Berriel, Cindy Chung, Oliver Xie, Angus Lewis, Calvin He, Cedomir Malgieri, Flint O'Neil, Janet Stefanov, and seminar participants at Stanford University for their helpful comments. This project was supported by the Dixon and Carol Doll Graduate Fellowship through the Stanford Institute for Economic Policy Research.

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A central question in international macroeconomics is the extent to which international financial markets allow countries to mitigate aggregate shocks. Despite substantial cross-border asset holdings, the standard view is not much: if investors were able to trade financial assets to fully share risk amongst themselves, then a country's consumption would decline relative to other countries when its exchange rate *appreciates*. This is inconsistent with the data, which show that a country's consumption declines when its exchange rate *depreciates*: this is the Backus and Smith (1993) puzzle. Full risk sharing therefore predicts the wrong correlation between exchange rates and the state of the economy. To make progress on this fundamental challenge for models of open economies, researchers have resorted to writing models that shut down most cross-border asset trade, so as to dramatically weaken international risk sharing. However, this reveals a deeper puzzle: how is it that in the world, cross-border asset trade is substantial, but exchange rate dynamics seem to require that the same asset trade be severely limited?

In this paper, I show that facts about foreign asset holdings discipline the amount of international risk sharing. To do so, I build a simple model of international business cycles with international financial frictions, where heterogeneous households must hold money to buy goods, with the key household-level friction being that repatriating foreign income to obtain money requires a costly financial intermediation. Once aggregated, my model is consistent with two key facts about foreign asset holdings: (i) home portfolio bias, which is the fraction of wealth that investors hold in domestic assets rather than foreign assets, and (ii) the elasticity of substitution of portfolio holdings across countries in response to expected returns, which I refer to as the *portfolio elasticity*. In the model, these two portfolio facts characterize how much international risk sharing occurs in equilibrium.

I show that a model with (i) financial frictions of the size needed to match the portfolio facts; and (ii) volatility in international relative demand for country-specific goods to match volatility in import expenditure shares suffices to quantitatively solve the Backus-Smith puzzle, and is consistent with two other important exchange rate patterns: exchange rates are much more volatile than most macroeconomic aggregates, and exchange rate fluctuations are persistent, with a half life of around 5 years (the *purchasing power parity* puzzle of Rogoff, 1996). Thus, observed exchange rate dynamics do not require severe limitations on international asset trade to arise in equilibrium.

Why does the Backus-Smith puzzle emerge with full risk sharing? In the international setting, different countries consume different goods, so full risk sharing equalizes not marginal utility across countries, but rather the marginal utility *per additional dollar*. Thus, if a country's consumption declines, so the marginal utility of its households increases, it can only be that the value of an additional dollar has declined for them. This means that its domestic price level, expressed in dollars, must have increased relative to other countries, which is a real exchange rate appreciation. But in the data, a decrease in a country's relative consumption is correlated with exchange rate depreciation.

In my model, a combination of home portfolio bias and shocks to the relative demand for country-specific goods resolves the Backus-Smith puzzle. When the relative demand for American goods increases, they become more expensive relative to foreign goods. American households, who mostly consume American goods, face a higher price level relative to foreign households, which is a US real exchange rate appreciation. American firms, which produce American goods, become more profitable and pay out more dividends to investors. With imperfect international financial markets, an increase in American dividends requires an increase in the return on American-originated securities relative to their foreign counterparts to clear financial markets. Under home portfolio bias, American households own mostly American securities, so their portfolio returns are higher, their portfolio income increases and they consume more. Hence, American households increase their relative consumption following a US real exchange rate appreciation, as in the data, resolving the Backus-Smith puzzle.

The key difference in my setting is that marginal utility per dollar is no longer equalized across countries: instead, the amount of international risk sharing, which I define as the cross-country co-movement of the marginal utility per dollar, is determined by the amount of home portfolio bias and the portfolio elasticity. Nevertheless, a quantification of the model suggests that marginal utility per dollar does still co-move strongly: the Backus-Smith puzzle is neither evidence that international risk sharing is especially weak, nor evidence that international financial markets fail to allow countries to mitigate aggregate shocks. Indeed, when relative demand for American goods increases, the fairly small international financial frictions that match the portfolio facts suffices to allow American households to increase their relative consumption.

In Section 1, I build on the international business cycle model of Backus, Kehoe and Kydland (1993). Households trade a complete set of contingent claims to dividend income (as in Arrow, 1964 and Debreu, 1959), capturing the wide variety of assets traded across countries. Following the international monetary model of Lucas (1982), I introduce money as a payment technology: consumers must buy goods with money. The model introduces a financial friction: consumers must procure costly financial services to turn dividend income into money. The need for financial services varies by household, who are therefore heterogeneous, and by the country the income originates from. The average household needs to use more financial services to repatriate foreign dividend income relative to domestic income, making foreign asset holdings less attractive. Section 2 shows that these financial frictions introduce home bias in aggregate country-level portfolios. Section 3 shows how aggregate portfolios characterize international risk sharing, and breaks down the model's mechanism for how home portfolio bias and relative demand shocks resolve the Backus-Smith puzzle. Conversely, productivity shocks raise the supply of a country's good and lower its price while increasing its consumption; hence, they continue to drive the opposite correlation between relative consumption and real exchange rate.

In Section 4, I document facts about international portfolios and international trade expenditures, and quantify the model to match those facts. Section 5 shows that both home portfolio bias and relative demand shocks are key to resolving the exchange rate puzzles. The model also matches classical business cycle moments. Then I apply a Kalman filter to reconstruct the time series of US productivity and relative demand shocks using business cycle data. In line with the business cycle literature, it shows that productivity shocks drive most GDP and consumption fluctuations, but relative demand shocks drive most exchange rate fluctuations. However, a counterfactual decomposition indicates that both productivity and relative demand shocks are needed to match the Backus-Smith correlation seen in the data.

Section 6 considers two other popular explanations for the Backus-Smith puzzle. I add intermediation shocks as in Itskhoki and Mukhin (2021) and discount factor shocks as in Kekre and Lenel (2024a) to the model, and re-run the Kalman filter to back out the time series of these shocks. The results suggest that there is little room for these shocks to explain exchange rate movements. This is because in my environment where foreign asset holdings are as in the data, there is substantially more international risk sharing than in their environments, so these shocks do not generate large exchange rate volatility.

Literature French and Poterba (1991) document that home portfolio bias persisted despite moves toward open financial markets in the 1980s, and consider various international financial frictions that might inhibit portfolio diversification. The large following literature is surveyed by Lewis (1999) and Coeurdacier and Rey (2013). One popular approach posits that holding home assets helps households hedge against various kinds of risk. For example, Pavlova and Rigobon (2007) show that home assets help households hedge against exchange rate risk caused by discount rate shocks: households have a high discount rate precisely when domestic goods are expensive.[1] The main difficulty faced by this set of models is that asset trade allows households to implement perfect international risk sharing, which is inconsistent with exchange rate facts.[2] Hence, returning to the approach of French and Poterba (1991), I let international financial frictions determine portfolio allocations.

This paper also develops the general equilibrium macroeconomic implications of the new literature on inelastic international financial markets following Koijen and Yogo (2020), Camanho, Hau and Rey (2022), and Jiang, Richmond and Zhang (2024b, 2025), who provide estimates of the portfolio elasticity. In this sense, it complements the work of Kleinman, Liu, Redding and Yogo (2023), who discuss its implications for convergence to steady state in the neoclassical growth model.

My treatment of the exchange rate puzzles has been influenced by a literature that documents mechanisms that unsatisfactory to resolve these puzzles. Chari, Kehoe and McGrattan (2002) show that monetary shocks and nominal price rigidities replicate neither the cyclicality nor the persistence of exchange rate fluctuations. They and Lustig and Verdelhan (2019) show that restricting international asset trade to only nominal risk-free bonds also does not help explain the cyclicality of exchange rate fluctuations.[3] This complements earlier work by Baxter and Crucini (1995), Kollmann (1996), Heathcote and

[1] Another approach posits that holding home equities helps households insure against non-insurable labor income risk. While Baxter and Jermann (1997) show that this approach fails in the simplest one-good setting, Heathcote and Perri (2013) show that this mechanism helps in economies with multiple goods and investment. Coeurdacier and Gourinchas (2016) show that adding international trade in bonds also overturns Baxter and Jermann's result.

[2] Say if there are N countries, and each country has just two assets that can be traded (a stock and a bond), and no asset is redundant. To avoid perfect international risk sharing, there must be risks that are not spanned by the asset payoffs, so the model needs at least $2N + 1$ shock processes. As the number of tradable assets increases, the required number of shocks becomes more demanding.

[3] Some recent work finds that a combination of domestic and international market incompleteness may resolve the puzzle, under certain assumptions about the correlation between idiosyncratic uninsurable household-level risk and aggregate consumption (Marin and Singh, 2025; Acharya, Challe and Coulibaly, 2025).

Perri (2002), and Kehoe and Perri (2002), who show that such restrictions do not improve on the fit of the international business cycle model of Backus, Kehoe and Kydland (1992). It also complements Cole and Obstfeld (1991), who show that asset trade is redundant with sufficient goods trade. Hence, my mechanism relies neither on nominal rigidities, nor on restrictions on the span of traded assets.

Models of international risk sharing and exchange rates can be roughly divided into three categories, depending upon their degree of international financial market segmentation. The first category of papers allow investors to trade a complete set of financial securities across countries. This approach is taken by many models which link exchange rates with a broader class of risky assets, such as Verdelhan (2010), Colacito and Croce (2011, 2013), and Colacito, Croce, Ho and Howard (2018), as the First Welfare Theorem makes computing allocations and asset prices particularly tractable. As investors implement full risk sharing when allowed to trade a complete set of securities, the positive correlation between relative consumption and exchange rate appreciations is the key puzzle for these papers. I show how to resolve the puzzle while retaining the tractability afforded by complete securities markets.

The second category features some degree of international financial frictions. Existing approaches include limited financial participation (Alvarez, Atkeson and Kehoe, 2002, 2009), portfolio adjustment costs (Fukui, Nakamura and Steinsson, 2023; Guo, Ottonello and Perez, 2023), and convenience yields (Jiang, Krishnamurthy and Lustig, 2023; Jiang, Krishnamurthy, Lustig and Sun, 2024a; Kekre and Lenel, 2024b). I show that these mechanisms can be disciplined by international portfolio facts.

A third category features a strong degree of segmentation between financial markets between countries. Corsetti, Dedola and Leduc (2008) make progress on the cyclicality puzzle with a low trade elasticity and restricting international asset trade to one-period bonds denominated in a global numéraire. A popular recent strand of literature follows Gabaix and Maggiori (2015): in these models, households and firms are limited to trading assets domestically; international asset trade is restricted to risk-averse financial intermediaries. In particular, Itskhoki and Mukhin (2021) and papers that borrow its mechanism, such as Kekre and Lenel (2024a), do resolve many of the exchange rate puzzles, but they ignore that foreign asset holdings allow countries to share risk: instead, their foreign portfolio shares are essentially zero.[4] I show that the exchange rate puzzles can be resolved in a model that is consistent with international portfolio facts and amenable to standard approaches to asset pricing.

1 Model

The world economy has countries $i = 1, \ldots, N$. Within each country lives a large number of infinitely-lived households and firms. In each country, the identical home firms own physical capital and produce a traded country-specific intermediate good by hiring labor from home households, as in the real business cycle model of Backus, Kehoe and Kydland (1992). Final goods producers combine intermediate goods from different countries to

[4]In Itskhoki and Mukhin (2021), financial intermediaries sell short-term bonds in one currency and buy short-term bonds in the other currency. There are no international equity holdings.

produce non-traded final goods that households consume and non-traded capital goods that firms invest in. The home intermediate good makes up a large share of home final goods and a small share of foreign final goods, reflecting home bias in consumption. A new feature in my setting is shocks to the relative demand for each country's intermediate good.

Each country has its own currency, in which all local prices are denominated. In particular, country i's final good costs P_i^F units of local currency. The nominal exchange rate $\mathcal{E}_{ij}$ is the price of a unit of country j's currency in country i's currency. More simply, it is the units of i's currency per unit of j's currency; a good that costs P units of j's currency equivalently costs $\mathcal{E}_{ij}P$ units of i's currency. A higher $\mathcal{E}_{ij}$ means that j's currency appreciates, and i's currency depreciates. The real exchange rate $\mathcal{Q}_{ij}$ is the relative price of each country's (non-traded) final good:

$$\mathcal{Q}_{ij} \equiv \frac{\mathcal{E}_{ij} P_j^F}{P_i^F}.$$

In the data, real and nominal exchange rates move close to one-for-one (see Appendix Figure 12), so in the model I have them move one-for-one by assuming that consumer price inflation is identical across countries.

1.1 Timing

The timing within each period t is a slight modification of the international monetary model of Lucas (1982). Before any actions are taken, the state of the world s^t is revealed. I use $s^t = (s_0, \ldots, s_t)$ to denote the history of events from the beginning of time through to t. The probability at time 0 of a particular history s^t is $\pi(s^t)$. The timing of a household's actions is illustrated in Figure 1.

At the beginning of the period, intermediate goods firms hire workers to produce output, which they sell to final goods producers on credit, and credit wages to workers and dividend payouts to investors. Next, a financial market opens in each country, where households trade the securities of that country and may take out dividend income from their investment position. Simultaneously, a foreign exchange market opens, where households and final goods producers arrange to exchange currencies. Settlement in these markets is delayed until the end of the period. After asset trading ends, households convert their promised wages and dividends into money, which is the payment technology used in the goods market.[5] I introduce a new financial friction: households need to use financial services to obtain money. After receiving money, households trade it for goods with final goods producers.

At the end of the period, settlement occurs in the intermediate goods market, each country's financial market, and foreign exchange market: final goods producers use their accumulated money to settle their transactions with intermediate firms, who in turn use the money to settle their wage and dividend obligations, which was used to back the money that households received. In this setup, money is used as a unit of value and as a medium of exchange, but not as a store of value: no money is held across periods,

[5] Lucas (1980) models why money may be an efficient payments technology for goods markets.

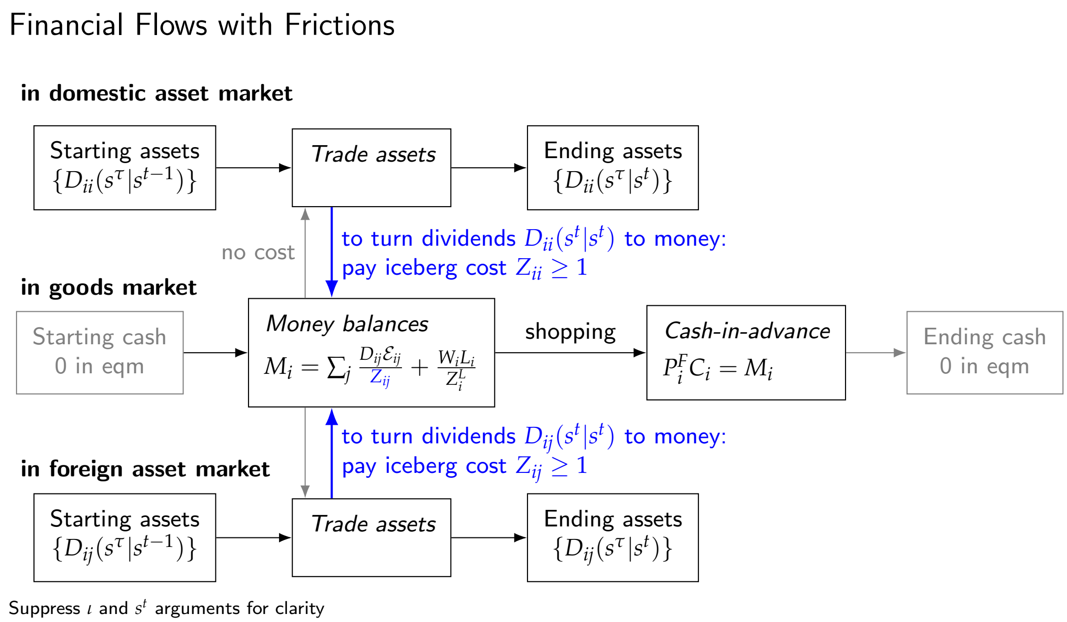

Figure 1: Household Financial Flows

and therefore money imposes no distortions on the real economy other than the financial services consumed in its creation.

1.2 Preferences and Technology

Let ι denote a household in country i. Household ι has utility function over its consumption of goods $C_i^g(\iota, s^t)$ and its labor supply $L_i(\iota, s^t)$

$$\sum_t \sum_{s^t} \beta^t \pi(s^t) u\big(C_i^g(\iota, s^t) - v(L_i(\iota, s^t))\big). \tag{1}$$

Flow utility $u(x) = (x^{1-\gamma} - 1)/(1-\gamma)$ is CRRA and the disutility of labor $v(L)$ follows Greenwood, Hercowitz and Huffman (1988, hereafter GHH) as being stated in consumption units, which will be convenient for proving an aggregation result later. GHH preferences increase the household's marginal utility of consumption when they supply more labor.

Each country's firms produce a country-specific traded intermediate good. A representative firm in country i enters the state s^t with capital $K_i(s^{t-1})$, whereupon it is subjected to a productivity shock to its technology $A_i(s^t)$. Technology has a global component a_{Gt} and a country-specific idiosyncratic component a_{Iit}:

$$\log A_i(s^t) \equiv a_{it} = \Gamma_i^a a_{Gt} + a_{Iit},$$

where Γ_i^a denotes the loading of country i on the global component a_{Gt}. The two processes

a_{Gt} and a_{Iit} are AR(1) with persistence ρ_a:

$$a_{Gt+1} = \rho_a a_{Gt} + \varepsilon_{Gt+1}$$
$$a_{Iit+1} = (1-\rho_a)\log \bar{A}_i + \rho_a a_{Iit} + \varepsilon^a_{Iit+1},$$

where $\bar{A}_i$ is country i's steady state productivity, and the innovations ε have mean zero. The firm hires labor $L_i(s^t)$ from i's households to operate its capital, thus producing a quantity

$$Y_i(s^t) = A_i(s^t) K_i(s^{t-1})^{\alpha} L_i(s^t)^{1-\alpha} \tag{2}$$

of the country i-specific intermediate good, which it sells at the competitive price $P_i^X(s^t)$.

Competitive final goods producers combine intermediates goods from various countries to produce non-traded final goods, as in Armington (1969). Engel (1999) and Chari, Kehoe and McGrattan (2002) provide the motivating evidence for an Armington setup instead of explicitly modelling tradable and non-tradable goods: the relative price of tradable and non-tradable goods are stable across time and do not fluctuate with the real exchange rate.[6]

Final goods producers in assemble intermediate goods from each country j into a non-traded country-specific final good that may be consumed directly by households, or transformed into financial services or units of physical capital. Country i's final goods producers have production function:

$$\left[\sum_j \eta_{ij}(s^t)^{\frac{1}{\theta}} X_{ij}(s^t)^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}}, \tag{3}$$

where $X_{ij}(s^t)$ is the usage of j's intermediate good, $\eta_{ij}(s^t)$ is the CES weight on j's intermediate good, and θ is the elasticity of substitution between each country's intermediate good, or the *trade elasticity* for short. The economy features home bias in consumption and capital investment, so the steady state weights satisfy $\bar{\eta}_{ii} > \bar{\eta}_{ij}$ for $j \neq i$, which Obstfeld and Rogoff (2000) argue to be an important element in fitting international economics facts.

I introduce shocks ω_{jt} to the weight of country j's intermediate good in producing each country i's final good as

$$\eta_{ij}(s^t) = \frac{\bar{\eta}_{ij}\exp(\omega_{jt})}{\sum_{\hat{j}} \bar{\eta}_{i\hat{j}}\exp(\omega_{\hat{j}t})}, \tag{4}$$

which I will show are a key feature of international trade data in Section 4.2. The denominator ensures that final goods are produced with constant returns to scale: $\sum_j \eta_{ij} = 1$. For convenience, I call ω_{jt} *relative demand shocks*, although they may be interpreted as increases in the factor-specific productivity of j's intermediates in producing final goods, or as a shock to consumer tastes for j's intermediates.[7] The shocks ω_{jt} follow the process

$$\omega_{jt+1} = \rho_\omega \omega_{jt} + \varepsilon^{\omega}_{jt}, \tag{5}$$

[6] This is different to the cross-country cross-section, where Balassa (1964) and Samuelson (1964) show that relative prices of tradable and non-tradable goods vary widely across countries.

[7] This interpretation is also consistent with a model where firms directly produce country-specific final goods, and households have CES preferences over each country's final good given by equation (3). The equations would be the same in such a model.

where the innovations $\varepsilon^{\omega}_{jt}$ have mean zero.

1.3 Household Portfolios and Money

Each country has a financial market in which households of all countries may trade a complete set of state-contingent nominal claims to each period's dividend payments by local firms. These claims may be thought of as state-contingent *dividend strips*.[8] In the state s^t, household ι owns $D_{ij}(\iota, s^\tau \mid s^t)$ dividend strips in country j, which pay out $D_{ij}(\iota, s^\tau \mid s^t)$ units of j's currency if the particular state s^τ occurs at time $\tau > t$, and 0 otherwise. Let $Q_j(s^\tau \mid s^t)$ denote the price of this security in units of currency j at time t. For there to be no arbitrage, the price of this security in time 0 is $Q_j(s^\tau) \equiv Q_j(s^\tau \mid s^0) = \prod_{t=0}^{\tau-1} Q_j(s_{t+1} \mid s^t)$.

In each country j's financial market, household ι may freely rebalance its portfolio subject to budget constraint

$$\begin{aligned} &D_{ij}(\iota, s^t \mid s^t) + \sum_{\tau>t}\sum_{s^\tau} Q_j(s^\tau \mid s^t) D_{ij}(\iota, s^\tau \mid s^t) \\ &\leq D_{ij}(\iota, s^t \mid s^{t-1}) + \sum_{\tau>t}\sum_{s^\tau} Q_j(s^\tau \mid s^t) D_{ij}(\iota, s^\tau \mid s^{t-1}). \end{aligned} \tag{6}$$

The household allows a quantity $D_{ij}(\iota, s^t) \equiv D_{ij}(\iota, s^t \mid s^t)$ of dividend strips to mature, and takes those units of j's currency to the foreign exchange market to obtain their own currency for consumption. At time 0, the household faces a budget constraint

$$\sum_t \sum_{s^t} \sum_j \mathcal{E}_{ij}(s^0) Q_j(s^t) D_{ij}(\iota, s^t \mid s^0) \leq \bar{B}_i(\iota) \tag{7}$$

that limits the value of its initial portfolio holdings by its endowment $\bar{B}_i(\iota)$, which is specified in country i's currency. The no-Ponzi scheme constraint is $\lim_{t\to\infty} D_{ij}(\iota, s^\tau \mid s^t) \geq 0$ for all $\tau > t$.

Households must hold money $M_i(\iota, s^t)$ in order to buy goods, imposing a *cash-in-advance* constraint:

$$P_i^F(s^t) C_i^g(\iota, s^t) \leq M_i(\iota, s^t), \tag{8}$$

where $P_i^F(s^t)$ is the price of the final good consumed by country i's households (denominated in i's currency) and $C_i^g(\iota, s^t)$ is household ι's consumption of goods. I depart from the literature by distinguishing consumption of goods, which directly contribute to household utility, from household consumption of financial services, which do not. Instead, financial services comprise all of the services that sit between a household's receipt of income and a household's receipt of goods: bank accounts, interest rate spreads on loans and mortgages, payment services (e.g. credit cards), insurance, etc.

I model financial services as an iceberg cost that households pay to obtain money from their dividend and wage income. For dividend income originating in country j, the household ι must expend a fraction $1 - 1/Z_{ij}(\iota)$ of the income on financial services,

[8] *Strips* (Separate Trading of Registered Interest and Principal Securities) break down a financial security into its cashflows and allow investors to trade those cashflows separately.

which are a transformation of the final good,[9] and receives the remaining fraction $1/Z_{ij}(\iota)$ as money. Hence, a household which takes out $D_{ij}(\iota, s^t)$ in dividend income from each country j in state s^t obtains the amount of money in currency i

$$\sum_j \frac{\mathcal{E}_{ij}(s^t) D_{ij}(\iota, s^t)}{Z_{ij}(\iota)}.$$

Households are heterogeneous in their $Z_{ij}(\iota)$, but these costs do not fluctuate across time.[10] For the average household ι, $Z_{ij}(\iota)$ will be larger than $Z_{ii}(\iota)$, so the household optimally chooses to hold a home-biased portfolio of dividend strips. Allowing investors to trade portfolios costlessly and imposing the cost on when the investment position is closed out makes the model substantially more tractable than imposing frictions on trading assets (e.g. Bacchetta and van Wincoop, 2010).

Household ι supplies $L_i(\iota, s^t)$ units of labor, and under the prevailing wage of $W_i(s^t)$, it receives labor income of $W_i(s^t) L_i(\iota, s^t)$ units of i's currency. Country i households similarly expend a fraction $1 - 1/Z_i^L$ of their labor income on financial services in order to convert it into money. Household ι's money balances comprise its dividend and labor income net of financial services costs, and unspent balance from the previous period:[11]

$$\begin{aligned} M_i(\iota, s^t) = \sum_j \left[\mathbf{1}_{D_{ij}(\iota, s^t) > 0} \frac{\mathcal{E}_{ij}(s^t) D_{ij}(\iota, s^t)}{Z_{ij}(\iota)} + \mathbf{1}_{D_{ij}(\iota, s^t) < 0} \mathcal{E}_{ij}(s^t) D_{ij}(\iota, s^t) \right] + \frac{W_i(s^t) L_i(\iota, s^t)}{Z_i^L} \\ + \left[M_i(\iota, s^{t-1}) - P_i^F(s^{t-1}) C_i(\iota, s^{t-1}) \right]. \end{aligned} \tag{9}$$

Proceeding under the usual assumption that monetary policy keeps the nominal interest rate positive, households do not carry money balances across periods. Thus, using the law of motion of money balances (9), the cash-in-advance constraint (8) can be written independently of money balances as

$$P_i^F(s^t) C_i^g(\iota, s^t) = \sum_j \left[\mathbf{1}_{D_{ij}(\iota, s^t) > 0} \frac{\mathcal{E}_{ij}(s^t) D_{ij}(\iota, s^t)}{Z_{ij}(\iota)} + \mathbf{1}_{D_{ij}(\iota, s^t) < 0} \mathcal{E}_{ij}(s^t) D_{ij}(\iota, s^t) \right] + \frac{W_i(s^t) L_i(\iota, s^t)}{Z_i^L}. \tag{10}$$

1.4 Household Problem

Household ι's problem is to choose paths of consumption of goods $C_i^g(\iota, s^t)$, labor $L_i(\iota, s^t)$, and asset portfolio $D_{ij}(\iota, s^\tau \mid s^t)$ to maximize its expected utility

$$\sum_t \sum_{s^t} \beta^t \pi(s^t) u\big(C_i^g(\iota, s^t) - v(L_i(\iota, s^t))\big)$$

[9] One may think of there being competitive financial intermediaries who operate a technology that converts final goods into financial services with constant returns to scale.

[10] Heterogeneity in $Z_{ij}(\iota)$ represents differing financial sophistication across households. More sophisticated investors repatriate foreign income more easily, as they are better equipped to deal with foreign exchange transaction and hedging costs, foreign banking systems, foreign tax withholding, and so forth.

[11] I assume that going short on dividend strips does not incur financial services costs, but I weaken this assumption slightly in Section 2.

subject to budget constraints (6) and (7), the no-Ponzi scheme constraint, and the cash-in-advance constraint (10). Household ι's first-order condition for labor equates the disutility of labor with the real wage:

$$v'(L_i(\iota, s^t)) = \frac{W_i(s^t)}{P_i^F(s^t) Z_i^L}. \tag{11}$$

I defer discussion of the household's portfolio problem to Section 2.

1.5 Firm Problem

The firm faces the standard problem of a neoclassical firm who owns capital: it chooses its labor demand to maximize its operating income

$$R_i^K(s^t) K_i(s^{t-1}) \equiv \max_{L_i(s^t)} \Big\{ P_i^X(s^t) A_i(s^t) K_i(s^{t-1})^{\alpha} L_i(s^t)^{1-\alpha} - W_i(s^t) L_i(s^t) \Big\}.$$

The firm's first-order condition for labor is

$$W_i(s^t) = (1-\alpha) P_i^X(s^t) A_i(s^t) \left(\frac{K_i(s^{t-1})}{L_i(s^t)} \right)^{\alpha}. \tag{12}$$

The firm's labor demand and operating income are, respectively,

$$L_i(s^t) = \left[(1-\alpha) A_i(s^t) \frac{P_i^X(s^t)}{W_i(s^t)} \right]^{\frac{1}{\alpha}} K_i(s^{t-1}). \tag{13}$$

$$R_i^K(s^t) K_i(s^{t-1}) = \frac{\alpha}{1-\alpha} W_i(s^t) L_i(s^t). \tag{14}$$

The firm chooses to invest an amount $I_i^K(s^t)$ in new capital, which it purchases at price $P_i^K(s^t)$. The firm pays out its remaining free cash flow $R_i^K(s^t) K_i(s^{t-1}) - P_i^K(s^t) I_i^K(s^t)$ to holders of dividend strips.

The firm chooses a path of investment $I_i^K(s^t)$ to maximize the present value of its dividend payouts[12]

$$\sum_{t=0}^{\infty} \sum_{s^t} Q_i(s^t) \big[R_i^K(s^t) K_i(s^{t-1}) - P_i^K(s^t) I_i^K(s^t) \big], \tag{15}$$

subject to the law of motion of capital

$$K_i(s^t) = (1-\delta) K_i(s^{t-1}) + I_i^K(s^t).$$

The firm's Euler equation for capital is

$$Q_i(s^t) P_i^K(s^t) = \sum_{s_{t+1}} Q_i(s^{t+1}) \big[R_i^K(s^{t+1}) + (1-\delta) P_i^K(s^{t+1}) \big]. \tag{16}$$

[12] Diamond (1967) shows that with trade in a complete set of contingent claims, the firm's choice of discount factor does not affect its behaviour. The firm's intertemporal objective function can be microfounded as maximizing the wealth of the firm's owners at time 0.

1.6 Final Goods Producer Problem

The final goods producer's problem is also standard: it chooses intermediates demand $X_{ij}(s^t)$ to maximize profits

$$P_i^F(s^t)\left[\sum_j \eta_{ij}(s^t)^{\frac{1}{\theta}} X_{ij}(s^t)^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}} - \sum_j \mathcal{E}_{ij}(s^t)P_j^X(s^t)X_{ij}(s^t), \tag{17}$$

which are zero in equilibrium, as final goods producers are competitive and face constant returns to scale. The first-order conditions can be stated in terms of i's total expenditure share on j's intermediate goods x_{ij}^{sh}:

$$x_{ij}^{sh} \equiv \frac{\mathcal{E}_{ij}P_j^X X_{ij}}{\sum_j \mathcal{E}_{ij}P_j^X X_{ij}} = \eta_{ij}\left(\frac{\mathcal{E}_{ij}P_j^X}{P_i^F}\right)^{1-\theta}. \tag{18}$$

1.7 Equilibrium

Equilibrium requires various market clearing conditions, each of which hold in every period t and every state of the world s^t. In country i, I_i^K units of the final good are purchased by firms for capital investment, C_i^g units are consumed directly by households, and the remainder is transformed into financial services to also be consumed by households. The market clearing condition for country i's final goods is such that their production (from eqn. 3) satisfies i's consumption and investment:

$$C_i(s^t) + I_i^K(s^t) = \left[\sum_j \eta_{ij}(s^t)^{\frac{1}{\theta}} X_{ij}(s^t)^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}}. \tag{19}$$

Total household consumption $C_i(s^t)$ is

$$C_i(s^t) \equiv \int C_i(\iota, s^t)d\iota = \int C_i^g(\iota, s^t) + \left[\frac{\sum_j \mathcal{E}_{ij}(s^t)D_{ij}(\iota, s^t) + W_i(s^t)L_i(\iota, s^t)}{P_i^F(s^t)} - C_i^g(\iota, s^t)\right]d\iota, \tag{20}$$

where the bracketed term is total financial services consumption. Market clearing for intermediate goods made by country j is

$$\sum_{i=1}^N X_{ij}(s^t) = A_j(s^t)K_j(s^t)^\alpha L_j(s^t)^{1-\alpha} \quad \text{for all } j = 1, \ldots, N, \tag{21}$$

Intermediate goods follows the law of one price: in each country i, j's good is priced at $\mathcal{E}_{ij}(s^t)P_j^X(s^t)$ in units of i's currency.[13] Labor market clearing equates labor demand with labor supply:

$$L_j(s^t) = \int L_j(\iota, s^t)d\iota \quad \text{for all } j = 1, \ldots, N.$$

[13]Under Armington trade, shipping costs, tariffs, and other importer-specific costs are absorbed into the input weight matrix η^F. For more details, see Section A.4.

Securities market clearing equates the total holdings of country j-originated dividend strips to their supply by country j's firms:

$$\sum_{i=1}^{N} D_{ij}(s^t) = \sum_{i=1}^{N} \int D_{ij}(\iota, s^t) d\iota = R_j^K(s^t) K_j(s^{t-1}) - P_j^K(s^t) I_j^K(s^t) \quad \text{for all } j = 1, \ldots, N. \tag{22}$$

Foreign exchange market clearing, which is the international balance of payments, equates the total demand for i's currency by importers and repatriation of foreign dividends with the total supply of i's currency from exporters and dividend payouts to foreign investors:

$$\begin{aligned} P_i^X(s^t) \sum_{j \neq i} X_{ji}(s^t) + \sum_j \int \mathcal{E}_{ij}(s^t) D_{ij}(\iota, s^t) d\iota \\ = \sum_{j \neq i} \mathcal{E}_{ij}(s^t) P_j^X(s^t) X_{ij}(s^t) + \sum_j \int D_{ji}(\iota, s^t) d\iota. \end{aligned} \tag{23}$$

Finally, the model maps real and nominal variables so that consumer price inflation $P_i^F(s^{t+1})/P_i^F(s^t)$ is constant across countries. Thus, real and nominal exchange rates co-move one-for-one.

An equilibrium has allocations for households $D_{ij}(\iota, s^\tau \mid s^t)$, $L_i(\iota, s^t)$, $C_i(\iota, s^t)$, $C_i^g(\iota, s^t)$; allocations for firms $L_i(s^t)$, $Y_i^X(s^t)$, $I_i^K(s^t)$; allocations for final goods producers $X_{ij}(s^t)$, $C_i(s^t)$, $I_i^K(s^t)$; goods prices $P_i^X(s^t)$, $P_i^F(s^t)$, $P_i^K(s^t)$; wages $W_i(s^t)$; and securities prices $Q_i(0, s^t)$ that satisfy: (i) household allocations solve their problem, (ii) firm allocations solve their problem, (iii) final goods producer allocations solve their problem, and (iv) the market-clearing conditions hold.

2 Portfolio Choice

In this section, I will compute the aggregate country-level portfolios. The financial services cost $Z_{ij}(\iota)$ are distributed according to a Fréchet (Type II extreme value) distribution:

$$\frac{1}{Z_{ij}(\iota)} \sim \text{Fréchet}\left(\frac{\kappa_0}{Z_{ij}}, \zeta - 1\right), \tag{24}$$

where the constant $\kappa_0 = \left(\frac{\Gamma(\zeta/(\zeta-1))}{\Gamma((\zeta-1/\gamma)/(\zeta-1))}\right)^\gamma$. The distribution of $Z_{ij}(\iota)$ across households is controlled by the scale parameter Z_{ij} and shape parameter ζ. Z_{ij} measures the cost faced by an average investor from country i when repatriating dividend income from country j (although the marginal investor in country j will tend to have a lower idiosyncratic draw of $Z_{ij}(\iota)$).[14] ζ measures the dispersion in the cost of realizing capital income across households, with larger ζ indicating smaller dispersion.

Household ι's portfolio composition is determined by its idiosyncratic draw of $Z_{ij}(\iota)$ for $j = 1, \ldots, N$: as the cash-in-advance constraint (10) is linear in dividend income

[14] Under the Fréchet distribution, a small number of households will draw $Z < 1$. For those households, I will impose a sufficiently large cost of moving money out of the goods market into the securities market to foreclose the possibility of a "money pump."

$D_{ij}(\iota, s^t)$ (assuming the household is not taking any short positions), in each state s^t household ι chooses its portfolio position such that it only holds dividend strips from the country with the highest expected return after adjusting exchange rates and the friction Z:

$$j(\iota, s^t) = \arg\max_j \frac{R^D_{ij}(s^t)}{Z_{ij}(\iota)}, \tag{25}$$

where $R^D_{ij}(s^t)$ is the exchange-rate adjusted return:

$$R^D_{ij}(s^t) = \pi(s^t)\frac{\mathcal{E}_{ij}(s^t)}{\mathcal{E}_{ij}(s^0)Q_j(s^t)}. \tag{26}$$

Consider household ι choosing between investing in the dividend strips of country i and j. The decision rule is a cutoff at $R^D_{ij}(s^t)/Z_{ij}(\iota) = R^D_{ii}(s^t)/Z_{ii}(\iota)$, or equivalently, $Z_{ij}(\iota)/Z_{ii}(\iota) = R^D_{ij}(s^t)/R^D_{ii}(s^t)$. Households ι with $Z_{ij}(\iota)/Z_{ii}(\iota)$ below this threshold invest in country j, and above this threshold invest in country i.

In Figure 2, I plot the density of the distribution of $Z_{ij}(\iota)/Z_{ii}(\iota)$ across households and illustrate how changes in the parameters of the distribution Z_{ij} and ζ shift portfolio allocations. The top row illustrates a state s^t where $R^D_{ij}(s^t) = R^D_{ii}(s^t)$, so the cutoff is at $Z_{ij}(\iota)/Z_{ii}(\iota) = 1$. As the Fréchet scale parameter Z_{ij} increases (moving from the top left to the top right panel), the mass of the distribution moves towards the right, so a smaller fraction of i's households fall below the cutoff where they invest in country j (shaded region). This makes the aggregate country-i portfolio more home-biased. Intuitively, the scale parameters Z_{ij} determine the degree of home portfolio bias.

As the Fréchet shape parameter ζ increases (moving from the bottom left to the bottom right panel), the dispersion in the relative cost $Z_{ij}(\iota)/Z_{ii}(\iota)$ declines. Then, following a relative increase in country j's expected returns $R^D_{ij}(s^t)/R^D_{ii}(s^t)$ from 1 to 1.05, which shifts the cutoff, a greater mass of households switch from investing in country i to country j (shaded region). Intuitively, the shape parameter ζ determines how much aggregate portfolio allocations respond to changes in expected returns.

2.1 Aggregate Portfolios

To analytically characterize aggregate portfolios, I define some objects. The aggregate dividend income of country i from country j is $D_{ij}(s^t) = \int D_{ij}(\iota, s^t)d\iota$. The aggregate of all dividend income of country i from every country as

$$D^{agg}_i(s^t) \equiv \left[\sum_j \left(\frac{\mathcal{E}_{ij}(s^t)D_{ij}(s^t)}{Z_{ij}}\right)^{\frac{\zeta-1}{\zeta}}\right]^{\frac{\zeta}{\zeta-1}}. \tag{27}$$

Country i's portfolio share in country j's dividend strips is therefore

$$d^{sh}_{ij}(s^t) = \frac{\mathcal{E}_{ij}(s^t)D_{ij}(s^t)/Z_{ij}}{D^{agg}_i(s^t)}. \tag{28}$$

Define country i's exchange rate and financial friction–adjusted aggregate household portfolio return R^H_i as

$$R^H_i(s^t)^{-1} = \sum_j \left[d^{sh}_{ij}(s^t) \times \left[R^D_{ij}(s^t)/Z_{ij}\right]^{-1}\right] \tag{29}$$

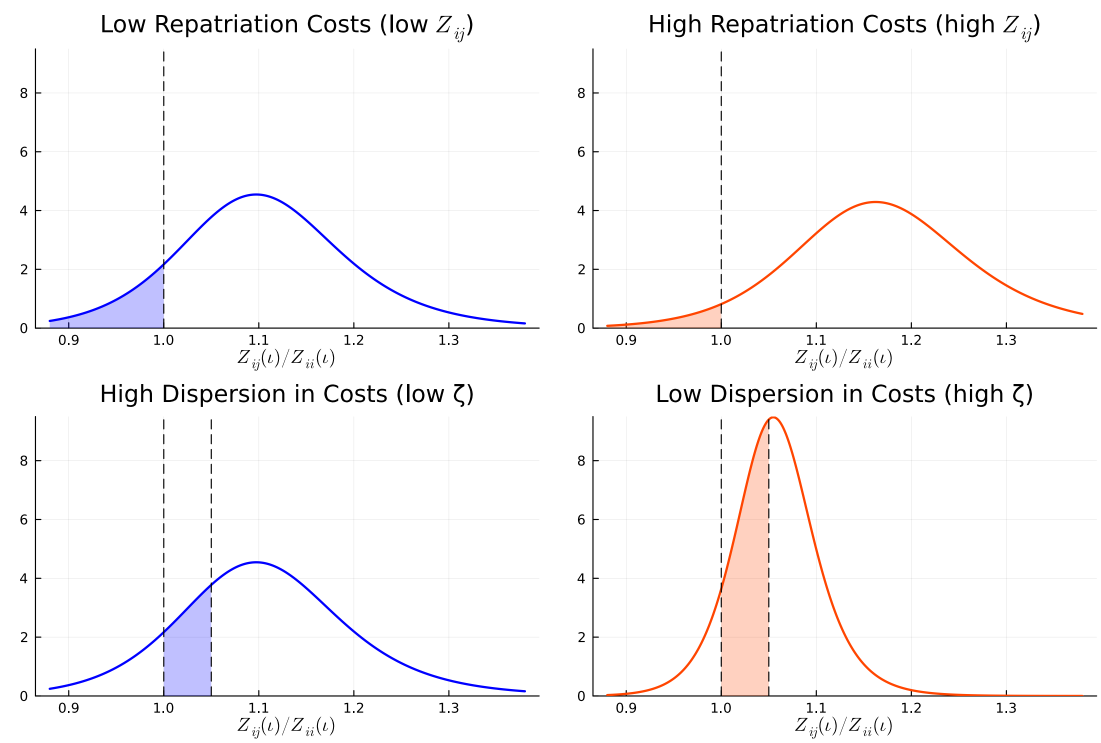

Figure 2: Density of Relative Financial Services Cost $Z_{ij}(\iota)/Z_{ii}(\iota)$ and Cutoff Rules for Household Portfolio Choice

Notes: Top row: portfolio allocation cutoffs with $R^D_{ij}(s^t) = R^D_{ii}(s^t)$: the cutoff is at $Z_{ij}(\iota)/Z_{ii}(\iota) = 1$, with households below the cutoff investing in country j (shaded region). Top right: as Z_{ij} increases, the mass of households below the cutoff decreases, so in aggregate, households become more home biased. Bottom row: change in portfolio allocation cutoffs when $R^D_{ij}(s^t)/R^D_{ii}(s^t)$ moves from 1 to 1.05. Bottom right: as ζ increases, the distribution of the relative financial services cost $Z_{ij}(\iota)/Z_{ii}(\iota)$ becomes less disperse, so following an in j's expected returns $R^D_{ij}(s^t)$ relative to $R^D_{ii}(s^t)$, more households shift portfolio allocation from country i to j.

Finally, assume that initial household wealth is distributed such that all households in country i have the same marginal utility of wealth μ_i. Proposition 1 shows how the portfolio shares of country i shift in response to changes in relative returns.

Proposition 1 (Portfolio demand)**.** *Country i's aggregate portfolio share in state-s^t dividend strips from country j is*

$$d_{ij}^{sh}(s^t) = \left(\frac{R_{ij}^D(s^t)/Z_{ij}}{R_i^H(s^t)}\right)^\zeta, \tag{30}$$

where i's aggregate portfolio return is

$$R_i^H(s^t) = \left(\sum_j \left[R_{ij}^D(s^t)/Z_{ij}\right]^{\zeta-1}\right)^{\frac{1}{\zeta-1}}. \tag{31}$$

Proposition 1 extends the framework of McFadden (1974) to a setting with a complete set of contingent securities, risk averse households, and endogenous labor income. The proof appears in Appendix A.1. The assumption of GHH preferences is key to obtaining this result, as labor income would otherwise fluctuate with returns, which would change households' intertemporal portfolio allocations to be inconsistent with equation (30).

Proposition 1 confirms the intuition in Figure 2: an increase in Z_{ij} increases the average household's cost of investing in country j, which reduces their portfolio share in j's dividend strips. The elasticity of the aggregate portfolio share in country j's dividend strips with respect to their relative return, or the *portfolio elasticity* for short, is

$$\frac{\partial \log d_{ij}^{sh}(s^t)}{\partial \log \frac{R_{ij}^D(s^t)/Z_{ij}}{R_i^H(s^t)}} = \zeta.$$

Proposition 2 shows that the heterogeneous household economy has a representative household representation.

Proposition 2 (Aggregation)**.** *Given prices, country i's aggregate consumption $C_i(s^t)$, labor $L_i(s^t)$, and portfolio $D_{ij}(s^\tau \mid s^t)$ are identical to those of the following representative household economy: the representative household chooses consumption of goods $C_i^g(s^t)$, labor $L_i(s^t)$, and portfolio $B_{ij}(s^t)$ to maximize*

$$\sum_t \sum_{s^t} \beta^t \pi(s^t) u\big(C_i^g(s^t) - v(L_i(s^t))\big) \tag{32}$$

subject to cash-in-advance constraint

$$P_i^F(s^t) C_i^g(s^t) = \kappa_1 D_i^{agg}(s^t) + \frac{W_i(s^t) L_i(s^t)}{Z_i^L}, \tag{33}$$

where $\kappa_1 = \kappa_0 / \Gamma(\frac{\zeta}{\zeta-1}) \approx 1$ and the dividend income aggregator D_i^{agg} is defined in equation (27), subject to a sequence of budget constraints in each country j's financial market

$$D_{ij}(s^t \mid s^t) + \sum_{\tau>t} \sum_{s^\tau} Q_j(s^\tau \mid s^t) D_{ij}(s^\tau \mid s^t) \leq D_{ij}(s^t \mid s^{t-1}) + \sum_{\tau>t} \sum_{s^\tau} Q_j(s^\tau \mid s^t) D_{ij}(s^\tau \mid s^{t-1}), \tag{34}$$

time-0 budget constraint

$$\sum_t \sum_{s^t} \sum_j \mathcal{E}_{ij}(s^0) Q_j(s^t) D_{ij}(s^t \mid s^0) \leq \bar{B}_i(\iota) \tag{35}$$

where the initial endowment is $\bar{B}_i = \int \bar{B}_i(\iota) d\iota$, and the no-Ponzi scheme constraint $\lim_{t\to\infty} D_{ij}(s^\tau \mid s^t) \geq 0$ for all $\tau > T$ in each j.

Equilibrium prices clear markets in the heterogeneous household economy. In the representative agent economy, Proposition 2 shows that the allocations are the same, so the same prices clear markets in that economy too (as firms are unchanged). Hence, equilibrium coincides across the two economies, and I use the representative agent economy to characterize the equilibrium going forward. The proof appears in Appendix A.2. I show that the representative household of country i's first-order condition for labor is the same as that of the heterogeneous households:

$$v'(L_i(s^t)) = \frac{1}{Z_i^L} \frac{W_i(s^t)}{P_i^F(s^t)}. \tag{36}$$

The assumption of GHH preferences is key to obtaining this result. I then show the representative household's portfolio is the same as the aggregate country-level portfolios of the heterogeneous households.[15]

The representative household i's first-order condition for country j's dividend strips equates the expected marginal utility of the consumption funded by an additional unit of j's dividends to the marginal utility of the wealth needed to buy the security:

$$\beta^t \pi(s^t) u'\big(C_i^g(s^t) - v(L_i(s^t))\big) \left[d_{ij}^{sh}(s^t)\right]^{-\frac{1}{\zeta}} \frac{\mathcal{E}_{ij}(s^t)}{P_i^F(s^t) Z_{ij}} = \mu_i \mathcal{E}_{ij}(s^0) Q_j(s^t), \tag{37}$$

where μ_i denotes the Lagrange multiplier on the time-0 budget constraint (35). I drop the constant κ_1 because it is very close to 1 for all values of ζ that I will consider. As a corollary to Proposition 1, substituting the portfolio share (30) into the first-order condition (37) obtains a simple expression for marginal utility:

$$\beta^t \pi(s^t) \frac{u'\big(C_i^g(s^t) - v(L_i(s^t))\big)}{P_i^F(s^t)} R_i^H(s^t) = \mu_i. \tag{38}$$

Hence, states of the world with high portfolio returns for i's households are states with low marginal utility for i's households.

[15] The representative household consumes a slightly different amount of goods $C_i^g(s^t)$ from the aggregate of the heterogeneous households, but this affects neither prices nor any other allocation in the model.

3 Mechanism: International Risk Sharing and the Exchange Rate Puzzles

Before quantifying the model, I describe how its mechanism relates to the various exchange rate puzzles identified in the literature.

3.1 International Risk Sharing

In an environment where households in different countries consume different goods, the relevant measure of amount of risk sharing is the co-movement in the marginal utility *per dollar*

$$\operatorname{corr}\left(\Delta \log\left(\frac{u_j'(s^t)}{\mathcal{E}_{ij}(s^t)P_j^F(s^t)}\right),\ \Delta \log\left(\frac{u_i'(s^t)}{P_i^F(s^t)}\right)\right).$$

When a household's consumption bundle is expensive, its marginal utility of having an additional dollar is low, so the additional dollar's marginal utility may be *higher* to a household with a cheap consumption bundle, even if that household has a *lower* marginal utility of consumption.

The international risk sharing condition follows from dividing the portfolio first-order condition (37) by the analogous condition with $j = i$, to obtain

$$\frac{\frac{u_j'(s^t)}{\mathcal{E}_{ij}(s^t)P_j^F(s^t)}}{\frac{u_i'(s^t)}{P_i^F(s^t)}} = \underbrace{\frac{\mu_j Z_{jj}}{\mu_i \mathcal{E}_{ij}(s^0) Z_{ij}}}_{\text{constant}} \times \left[\frac{d_{jj}^{sh}(s^t)}{d_{ij}^{sh}(s^t)}\right]^{\frac{1}{\zeta}}. \tag{39}$$

From this equation, Proposition 3 immediately follows.

Proposition 3 (Sufficient statistics)**.** *The amount of international risk sharing between country i and j, which measures co-movements in their relative marginal utility per dollar, is characterized by two sufficient statistics:*
(i) portfolio shares $d_{ij}^{sh}(s^t)$ and $d_{jj}^{sh}(s^t)$, and
(ii) the portfolio elasticity ζ.

This result, which is analogous to the celebrated result of Arkolakis, Costinot and Rodríguez-Clare (2012) in international trade, shows that household portfolios are key to international risk sharing. By re-arranging the marginal utility per dollar, the risk sharing condition (39) links real exchange rates with relative marginal utility across countries.

$$\frac{u_j'(s^t)}{u_i'(s^t)} = \underbrace{\frac{\mu_j Z_{jj}}{\mu_i \mathcal{E}_{ij}(s^0) Z_{ij}}}_{\text{constant}} \times \left[\frac{d_{jj}^{sh}(s^t)}{d_{ij}^{sh}(s^t)}\right]^{\frac{1}{\zeta}} \times \underbrace{\frac{\mathcal{E}_{ij}(s^t)P_j^F(s^t)}{P_i^F(s^t)}}_{\text{real exch. rate}}. \tag{40}$$

Equation (40) shows that international risk sharing is key to exchange rate dynamics.

3.2 Full Risk Sharing and the Backus-Smith Puzzle

In a model with no international financial frictions, dividend strips from different countries are perfect substitutes, and there is full risk sharing. My model collapses down to the this model when the cost of turning dividends to money is zero and portfolio holdings are perfectly elastic: $Z_{ij} = 1$ for all i, j and $\zeta \to \infty$. Substituting this into equation (39) shows that marginal utility per dollar co-moves perfectly. Substituting this into equation (40) obtains the international risk sharing condition of Backus and Smith (1993), which I state in the following proposition; I reproduce the original expression by removing the labor supply margin.

Proposition 4 (Backus and Smith, 1993)**.** *With $\zeta \to \infty$ and $Z_{ij} = 1$ for all i, j, households have perfect risk sharing: when country i's real exchange rate appreciates (the RHS of equation (41) declines), the marginal utility of i's households is high:*

$$\frac{u'(C_j(s^t))}{u'(C_i(s^t))} = \frac{\mu_j}{\mu_i \mathcal{E}_{ij}(s^0)} \underbrace{\frac{\mathcal{E}_{ij}(s^t) P_j^F(s^t)}{P_i^F(s^t)}}_{\text{real exch. rate}} \tag{41}$$

Hence, the consumption of i's households declines relative to j's households.

The Backus-Smith puzzle is that the full risk sharing condition (41) is violated in the data: when a country's real exchange rate appreciates, it is moderately correlated with *higher* relative consumption (with a magnitude of around 0.2). Proposition 4 shows that when households have the opportunity to trade securities and convert them into money costlessly, they implement full risk sharing: they fully smooth the marginal utility per dollar across countries; they only fail to completely smooth marginal utility because of fluctuations in the value of a dollar: final consumption goods are relatively more expensive in some states than others. i's households substitute consumption out of expensive states with high $P_i^F(s^t)$ relative to $\mathcal{E}_{ij}(s^t)P_j^F(s^t)$ toward states where consumption is cheap.

I emphasize three points about the risk sharing condition (41): first, it follows from the household block of the model alone; the precise nature of production is irrelevant. Second, the equation holds for *any* shock, so long as financial contracts can be written against it. Third, the assumption of trade in a complete set of contingent securities is not necessary for the qualitative direction of the correlation between exchange rates and marginal utility: it is sufficient for households to trade only one-period nominal risk-free bonds denominated in the various currencies, a very minimal market structure (Chari, Kehoe and McGrattan, 2002; Lustig and Verdelhan, 2019). For these reasons, the Backus-Smith puzzle has proven quite intractable.

With GHH preferences, Proposition 4 is less stark: movements in marginal utility are partly driven by fluctuations in labor supply. However, I will show quantitatively that GHH preferences alone do not resolve the puzzle.

3.3 Resolving the Backus-Smith Puzzle via Home Portfolio Bias

With home portfolio bias, following a real appreciation of currency i, the substitution effect towards states with cheap consumption outlined above is offset by changes in portfolio returns. Consider a relative demand shock $\omega_{it} > 0$ for country i's intermediate goods, which pushes up $\eta_{ji}(s^t)$ for all countries j (eqn. 4). All final goods producers increase their expenditure on i's intermediate good, but i's firms' marginal cost is upward sloping:

the capital stock of i's firms is fixed, and labor supply is upward sloping (households have increasing marginal disutility of work). Hence, its price P_i^X increases relative to that of foreign intermediates $\mathcal{E}_{ij}P_j^X$. This increases the relative price of country i's final goods

$$P_i^F = \left(\sum_j \eta_{ij} \left(\mathcal{E}_{ij} P_j^X \right)^{1-\theta} \right)^{\frac{1}{1-\theta}}$$

relative to that of foreign final goods, $\mathcal{E}_{ij}P_j^F$, as $\eta_{ii}^F > \eta_{ji}^F$ under home consumption bias. This is an appreciation in country i's real exchange rate.

The relative demand shock raises the prices received by i's firms, and they become more profitable, so $R_i^K K_i$ increases, and their total dividend payouts $R_i^K K_i - P_i^K I_i^K$ also increase (investment expenditure increases but not enough to fully offset the increase in flow profits). To clear the securities market (eqn. 22), an increase in the supply of i-originated dividend strips must be matched by an increase in total household portfolio holdings of i's dividend strips. Hence, the share of country j's portfolio invested in i's dividend strips in this state, which is d_{ji}^{sh}, must increase. The household on the margin between holding the dividend strips of country i and another country will choose i if the expected return on i's securities R_{ji}^D increases, as shown by the cutoff rules in the bottom row of Figure 2. The relationship between the portfolio share d_{ji}^{sh} and the expected returns on i's dividend strips R_{ji}^D is given by equation (30), reproduced below:

$$d_{ji}^{sh}(s^t) = \left(\frac{R_{ji}^D(s^t)/Z_{ji}}{R_j^H(s^t)} \right)^{\zeta}.$$

The portfolio elasticity ζ determines how much R_{ji}^D must increase for a given increase in d_{ji}^{sh}.

Under home portfolio bias, i's aggregate household portfolio puts a higher weight on i's dividend strips relative to foreign portfolios. Hence, an increase in expected returns on i's dividend strips increases the expected return on i's aggregate household portfolio R_i^H relative to the corresponding foreign return $R_j^H \mathcal{E}_{ij}(s^t)/\mathcal{E}_{ij}(s^0)$. This can be seen by noting that $d_{ii}^{sh} > d_{ji}^{sh}$ in the definition of R_j^H (eqn. 29), reproduced below:

$$R_j^H(s^t)^{-1} = \sum_j \left[d_{ji}^{sh}(s^t) \times \left[R_{ji}^D(s^t)/Z_{ji} \right]^{-1} \right].$$

An increase in i's portfolio returns increases i's total household dividend income relative to foreign households.

Proposition 5 shows the precise relation between the real exchange rate, marginal utility, and portfolio returns by substituting the portfolio demand equation (30) into the risk-sharing condition (eqn. 40).

Proposition 5 (Portfolio return effect)**.** *With financial frictions ($\zeta < \infty$ and $Z_{ij} \geq 1$),* *the international risk-sharing condition takes into account portfolio returns:*

$$\frac{u'_j(s^t)}{u'_i(s^t)} = \frac{\mu_j}{\mu_i \mathcal{E}_{ij}(s^0)} \times \frac{R_i^H(s^t)}{R_j^H(s^t)\frac{\mathcal{E}_{ij}(s^t)}{\mathcal{E}_{ij}(s^0)}} \times \underbrace{\frac{\mathcal{E}_{ij}(s^t) P_j^F(s^t)}{P_i^F(s^t)}}_{\text{real exch. rate}}. \tag{42}$$

Hence, the effect of i's real exchange rate appreciation on i's marginal utility u'_i is offset by a relative increase in i's household portfolio returns R_i^H. The change in marginal utility of i's households can be decomposed to first order as

$$du'_i(s^t) = \underbrace{\bar{u}''_i}_{<0} \times \left(\frac{\bar{C}_i^g}{\bar{C}_i^g - v(\bar{L}_i)} dC_i^g(s^t) - \frac{v(\bar{L}_i)}{\bar{C}_i^g - v(\bar{L}_i)} v'(\bar{L}_i) dL_i \right), \tag{43}$$

where bars denote the steady state of the variables. Following an increase in relative demand for i's goods, there is simultaneously an increase in i's labor supply L_i, which increases marginal utility u'_i, and an increase in i's consumption C_i^g, which decreases u'_i. Hence, i's representative household consumes relatively more than foreign households at the same time that i's real exchange rate appreciates, which resolves the Backus-Smith puzzle.

What fails without financial frictions? In that setting, while increases in i's firm profits raise dividend payouts by i's firms, they cannot increase the returns on i's state-contingent dividend strips in equilibrium, as households could profitably arbitrage any return differential. As there is no difference in portfolio returns across countries, i's households do not consume more.

3.4 Exchange Rate Volatility: Amplification of Demand Shocks

Home portfolio bias also amplifies the effect of demand shocks, which is necessary to match the volatility of real exchange rates. Recall that an increase in relative demand ω_{it} for i's goods increases the income and consumption of i's households. Under home consumption bias, an increase in the relative consumption of i's households increases the relative demand for i's goods. This amplifies the effect of the initial shock on total demand for i's goods, which further raises the relative profits of i's firms, which further increases the relative income of i's households. Hence, small demand shocks are amplified into large movements in relative prices and real exchange rates.

3.5 Exchange Rate Persistence: the Purchasing Power Parity Puzzle

The purchasing power parity puzzle is that fluctuations in real exchange rates are long-lived (Rogoff, 1996). With a shock that generates large fluctuations in real exchange rates, the puzzle that real exchange rate deviations are long-lived is solved by having this shock be persistent. This follows the argument in Engel and West (2005). I show in Section 4 that evidence from trade flows supports the notion that relative demand shocks are persistent.

With volatile shocks to the real exchange rate and slow mean reversion in these shocks, real exchange rates are difficult to forecast in the short-run. In this model, nominal

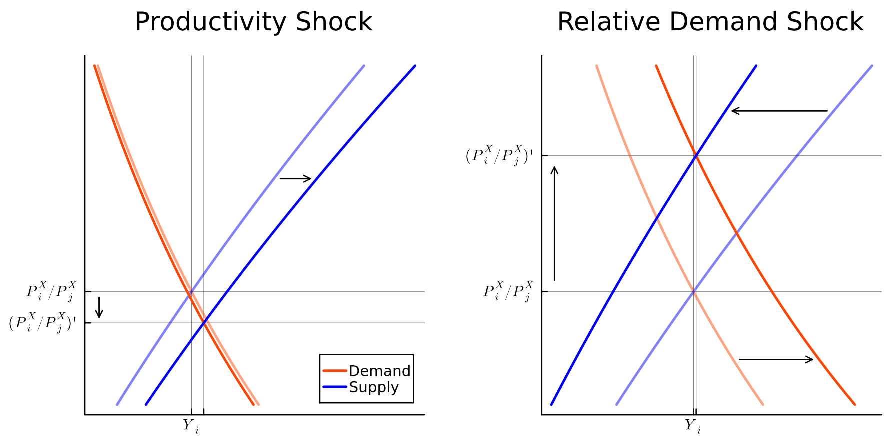

Figure 3: Supply and Demand: Productivity vs Relative Demand Shocks

exchange rate movements are perfectly correlated with real exchange rate movements, so they are also difficult to forecast, in line with the findings of Meese and Rogoff (1983).

3.6 Productivity vs Relative Demand Shocks

I introduce demand shocks because shocks to productivity A_i do not resolve the Backus-Smith puzzle, even with home portfolio bias and GHH preferences. This is most easily seen by looking at the demand and supply of i's intermediate goods: from the final goods producer's problem, demand is

$$\sum_j X_{ji} = \sum_j \eta_{ji} \left(\frac{\mathcal{E}_{ji} P_i^X}{P_j^F} \right)^{-\theta} \left[C_j + I_j^K \right]$$

and from the firm's problem, supply is

$$Y_i = A_i \left[(1-\alpha) A_i \frac{P_i^X}{W_i} \right]^{\frac{1-\alpha}{\alpha}} K_i.$$

In Figure 3, I plot the supply and demand curves for i's intermediates by varying $P_i^X / \mathcal{E}_{ij} P_j^X$ while holding other prices fixed. Then I plot the new supply and demand curves following shocks to productivity and relative demand, which I draw by holding other quantities and prices fixed at their post-shock values.

An increase in i's productivity (left panel) increases the supply of country i's intermediate goods, while demand is nearly unchanged, and hence its price P_i^X declines relative to foreign intermediates $\mathcal{E}_{ij} P_j^X$. Under home consumption bias, the price of country i's consumption bundle P_i^F declines relative to foreign bundle $\mathcal{E}_{ij} P_j^F$, so i's households choose to consume more. Hence, in response to a positive productivity shock, country i has a real depreciation at the same time its households increase their relative consumption.[16]

[16]This result is reminiscent of Cole and Obstfeld (1991), who show that in endowment economies, international trade in assets is redundant because movements in goods prices implement international risk sharing. In a production economy, productivity shocks are the closest analog to endowment shocks.

In contrast, an increase in relative demand for i's intermediate goods (right panel) shifts its demand curve outwards. At the same time, an increase in i's wages pushes the supply curve in. Hence, i's intermediates become relatively more expensive, and i's final good also becomes relatively more expensive, so i has a real appreciation.

Equilibrium in financial markets continues to be characterized by the risk-sharing condition (42). In response to a positive productivity shock to i's firms, competition pushes down the price of i's intermediate goods, resulting in minimal increases in the profits of i's firms. Hence, there is little relative movement in the dividends paid out by i's firms, so $R_i^H/(R_j^H \mathcal{E}_{ij})$ remains steady. The decline in the real exchange rate is realized in the international risk sharing equation (42) via a large increase in consumption overwhelming a smaller increase in labor supply.

4 Quantification

In this section, I quantify a two-country business cycle model. The two countries are the US and a composite of advanced economies that use the G10 currencies, which have freely floated against the US dollar since the end of the Bretton Woods system in 1973, and comprise most of the highly traded currencies in foreign exchange markets.[17]

Parameter values are summarized in Table 1. I describe the moments I use to estimate the parameters below.

4.1 Financial frictions

Financial frictions are parameterized by the portfolio elasticity ζ and international investment frictions Z_{ij}. Koijen and Yogo (2020) estimate an asset demand system covering short-term bonds, long-term bonds, and equities across countries. To get around endogeneity problems, they use a gravity equation to predict asset demand, and a regression of asset issuance on GDP and population to predict asset supply. Using their predicted asset demand and supply, they construct an instrumental variable for expected returns, and estimate that portfolio holdings in a particular asset class in a country respond to changes in expected returns with an elasticity of 15–25. I set the portfolio elasticity ζ to 21, in line with their estimates.

To quantify international financial frictions, I use the same measure of home portfolio bias as Coeurdacier and Rey (2013):

$$HB_{it} = 1 - \frac{\text{Share of Foreign Assets in } i\text{'s Portfolio}}{\text{Share of Foreign Assets in World Market Portfolio}}. \tag{44}$$

This measure is 0 when country i's portfolio treats domestic and foreign assets the same by holding them in the same ratio as their total supply, and is 1 when i's portfolio is entirely home-biased toward domestic assets. I plot these shares for the US equity

[17] The G10 currencies are the Euro post-1999, Japanese yen, British pound, Canadian dollar, Australian dollar, Swiss franc, Swedish krona, Norwegian krone, NZ dollar; pre-1999, I represent the Euro area by the Deutschmark (using West German data for the corresponding economic series), French franc, Italian lira, Spanish peseta, Dutch guilder, Belgian franc, Austrian schilling, Finnish markka. I construct the G10 sample by weighting each country/currency zone according to their nominal GDP in US dollars, converted at market rates. For data construction, see Appendix C.1.

Table 1: Quantification of Business Cycle Model

Parameter		Value	Moment	Target	Model
		Externally assigned			
γ	utility function curvature	2			
θ	trade elasticity	1.5	Feenstra et al., 2018		
ζ	portfolio elasticity	21	Koijen and Yogo, 2020		
		Endogenously chosen			
Production					
α	capital share	0.33	$\mathrm{E}[R_{it}^K K_{it}/P_{it}^X Y_{it}]$	0.33	0.33
δ	depreciation rate of capital	0.10	$\mathrm{E}[I_{it}^K/(C_{it}+I_{it}^K)]$	0.24	0.24
ρ_a	persistence of TFP shock	0.66	$\mathrm{corr}[a_{it}, a_{i,t-1}]$	0.47	0.45
σ_G	volatility of global shock	1.1%	$\mathrm{corr}[\Delta a_{it}, \Delta a_{jt}]$	0.50	0.51
σ_{Ia}	vol of idiosync. TFP shock	1.1%	$\sigma[\Delta a_{it}]$	1.2%	1.2%
Utility					
β	time preference	0.96	$\mathrm{E}[K_{it}/Y_{it}]$	2.5	2.3
χ_0	weight of labor in utility	4.95	$\mathrm{E}[L_{it}]$	0.20	0.20
χ_1	elasticity of labor supply	1.67	$\sigma[L_{it}]/\sigma[Y_{it}]$	0.79	0.66
International trade					
$\bar{\eta}_{ij}$	expenditure weight: imports	0.10	$1-\mathrm{E}[P_{it}^X X_{iit}/P_{it}^F(C_{it}+I_{it}^K)]$	0.10	0.10
ρ_ω	persistence of ω shock	0.85	$\mathrm{corr}[\omega_{ijt}, \omega_{ij,t-1}]$	0.85	0.85
$\sigma_{I\omega}$	vol of idiosync. ω shock	8.9%	$\sigma[\omega_{ijt}-\rho_\omega \omega_{ij,t-1}]$	7.1%	6.9%
International finance					
Z_{ii}	home financial friction	1.07	$\mathrm{E}[(C_{it}-C_{it}^g)/C_{it}]$	0.07	0.07
Z_{ij}	foreign financial friction	1.18	$\mathrm{E}[HB_{it}]$	0.76	0.75

Notes: all parameters and moments are stated in annual terms. The model is estimated at quarterly frequency. I normalize steady state TFP $\bar{A}_i = 1$, each country's TFP weighting on the global shock $\Gamma_i^a = 1$, and the marginal utility of wealth $\mu_i = 1$.

portfolio in Figure 4; they imply that the average home equity bias $\overline{HB}_i$ has been 76% in the post–Bretton Woods era. Coeurdacier and Rey (2013) and Coppola, Maggiori, Neiman and Schreger (2021) document similar home portfolio bias patterns across asset classes, including bonds, equities, asset-backed securities, and banking assets, and similar patterns across countries.[18]

To interpret the size of the international financial frictions Z, which are a one-off cost paid upon converting dividends into money, it is helpful to express them as a per-period wedge on gross returns, as in Jiang, Krishnamurthy and Lustig (2023). This calls for dividing $Z-1$ by the duration of household portfolios. In steady state, household portfolios are equivalent to a perpetual bond, for which the duration is simply

$$\mathit{Duration} = \frac{1}{1-\beta}.$$

The derivation is in Appendix A.3. With $\beta = 0.96$, the duration of household portfolios

[18]Hence, I take the level of home equity bias as representative of the ownership of claims to home dividend income. There may be discrepancies between the two concepts due multinational corporations, where firms listed on one stock exchange own capital in other countries, and from privately-held companies, whose equity are not generally available for portfolio investment and whose ownership is presumably more home-biased.

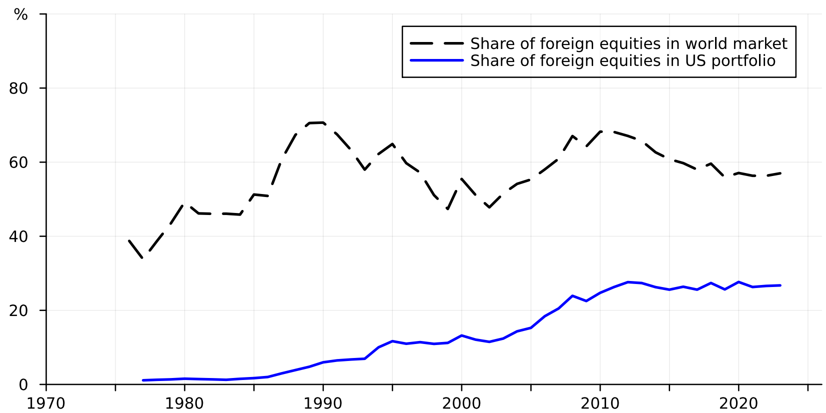

Figure 4: US Aggregate Equity Portfolio

Notes: data from International Financial Statistics (IMF, 2025) and World Federation of Exchanges (2025).

is 25 years.

I choose $Z_i^L = Z_{ii}$ and Z_{ij} so that $(Z_{ii}-1)/Duration = 28$bps and $(Z_{ij}-Z_{ii})/Duration =$ 44bps to match the average 7% US household consumption share on financial services and the average US home equity bias of 76% in the post–Bretton Woods era. Hence, to rationalize the observed pattern of home portfolio bias, it is as if the average investor faces an additional annual cost of 44bps of returns in holding foreign securities rather than domestic securities. This number is similar to the size of financial frictions suggested by Jiang, Krishnamurthy and Lustig (2023).[19]

4.2 International Trade and Relative Demand Shocks

I identify relative demand shocks using expenditure shares, following Dekle, Eaton and Kortum (2007) and Caliendo and Parro (2015). To do so, recall the expression for i's expenditure share on j's goods from equation (18):

$$x_{ijt}^{sh} \equiv \frac{\mathcal{E}_{ijt} P_{jt}^X X_{ijt}}{\sum_j \mathcal{E}_{ijt} P_{jt}^X X_{ijt}} = \frac{\mathcal{E}_{ijt} P_{jt}^X X_{ijt}}{P_{it}^F [C_{it} + I_{it}^K]} = \eta_{ijt} \left(\frac{\mathcal{E}_{ijt} P_{jt}^X}{P_{it}^F} \right)^{1-\theta},$$

where the second equality follows from market clearing for final goods. To calculate x_{ijt}^{sh}, I use import expenditure data $\mathcal{E}_{ijt} P_{jt}^X X_{ijt}$ from the IMF's Direction of Trade Statistics, which is available at an annual frequency, and calculate the denominator $P_{it}^F [C_{it} + I_{it}^K]$ as GDP less net exports at current prices $P_{it}^X Y_{it} - [P_{it}^X \sum_j X_{jit} - \sum_j \mathcal{E}_{ijt} P_{jt}^X X_{ijt}]$. I obtain the price index $\mathcal{E}_{ijt} P_{jt}^X / P_{it}^F$ by multiplying j's export price deflator by the bilateral exchange rate $\mathcal{E}_{ijt}$, and dividing by i's consumer price index. Finally, I set the elasticity of trade θ to 1.5, in line with evidence in Feenstra, Luck, Obstfeld and Russ (2018) and which is the conventional parameter in international macroeconomics.[20] By rearranging the

[19] This figure is also well within the range of management fees charged by mutual funds that actively trade foreign currencies, as catalogued in Bacchetta and van Wincoop (2010).

[20] Estimates of long-run trade elasticities tend to be larger, see e.g. Caliendo and Parro (2015).

Table 2: Expenditure Shares and Expenditure Weights

i	j		Autocorrelation	Volatility (%)	Correlation with $\Delta \log \mathcal{E}_{ijt}$
USA	G10	$\Delta \log x^{sh}_{ijt}$	0.78	7.3	0.17
		$\Delta \log \eta_{ijt}$	0.81	9.2	0.50
G10	USA	$\Delta \log x^{sh}_{ijt}$	0.74	8.0	0.55
		$\Delta \log \eta_{ijt}$	0.88	11.0	0.70
			ρ_ω	$\sigma[\varepsilon^\omega_t]$ (%)	
USA	G10	$\omega_j - \omega_i$	0.81	9.1	
G10	USA	$\omega_j - \omega_i$	0.88	11.1	

Notes: data 1973–2019, in annual terms. x^{sh}_{ijt} is country i's expenditure share on country j's goods, and η_{ijt} is country i's CES weight on country j's goods. The bilateral US-G10 exchange rate is the average of quarterly observations over the year; the G10 sample is described in Footnote 17.

equation for the expenditure share, I obtain the time series $\eta_{ijt} = x^{sh}_{ijt}/(\mathcal{E}_{ijt} P^X_{jt}/P^F_{it})^{1-\theta}$.

The key finding is that bilateral US-G10 expenditure shares x^{sh}_{ijt} and CES expenditure weights η_{ijt} are volatile, positively correlated with appreciations of j's exchange rate $\mathcal{E}_{ijt}$, and long-lived, as summarized in the upper panel of Table 2. An increase in expenditure on j's goods accompanying an appreciation of j's currency is consistent with the predictions of the model, and inconsistent with models that hold expenditure weights fixed: in those models, with the usual trade elasticity $\theta > 1$ (so that the Marshall-Lerner condition holds), an appreciation of j's currency would accompany a *decline* in expenditure on j's goods. The high persistence of shifts in expenditure shares, as seen in their high annual autocorrelations, strongly suggest that these patterns are not driven by pricing-to-market or other forms of nominal price rigidities: evidence suggests that international prices are reset at annual or greater frequency (Gopinath and Itskhoki, 2010).

Using data on η_{ijt}, I obtain a time series for ω_{jt} by inverting equation (4) to obtain $\omega_{jt} - \omega_{it} = \log((\eta_{ij}/\bar{\eta}_{ij})/(\eta_{ii}/\bar{\eta}_{ii}))$. I plot these time series in Figure 5, using US expenditure shares to recover ω, and using foreign expenditure shares. The time series co-move with each other, and with the US dollar exchange rate. These time series give me two ways to estimate the stochastic process of $\omega_{jt} - \omega_{it}$, which from equation (5) is

$$\omega_{jt+1} - \omega_{it+1} = \rho_\omega \big(\omega_{jt} - \omega_{it}\big) + \varepsilon^\omega_{t+1},$$

where $\varepsilon^\omega_{t+1} = \varepsilon^\omega_{jt+1} - \varepsilon^\omega_{it+1}$. I first estimate this equation using US import expenditure on goods from G10 economies, and then estimate vice versa, with the results in the bottom panel of Table 2. For a two-country symmetric quantification, I take the average of these two estimators as targets: $\rho_\omega = 0.85$ and $\sigma[\omega_{jt} - \omega_{it}] = \sqrt{2} \times \sigma_{I\omega} = 10.1\%$.

4.3 Other Parameters

Internationally-comparable direct measures of total factor productivity remain elusive. Hence, I follow the method of Backus, Kehoe and Kydland (1992) to back out the time series of productivity: for each country, I estimate total factor productivity as $\log A_{it} = \log Y_{it} - (1 - \alpha) \log L_{it}$ using aggregate data on output Y_{it} and employment

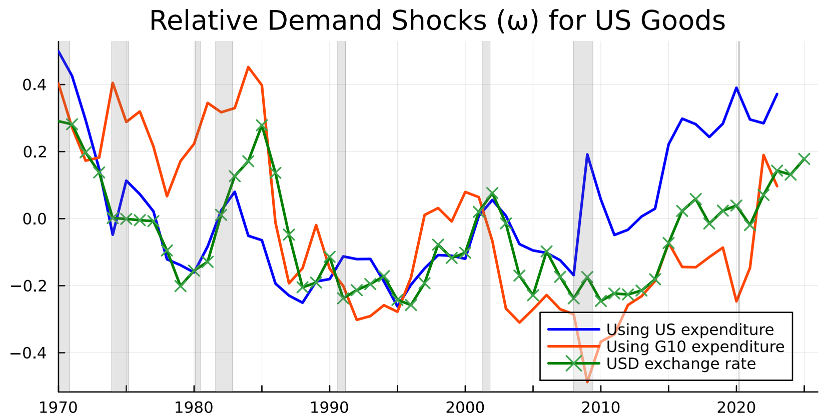

Figure 5: Relative Demand Shocks for US Goods
See text of Section 4.2 for construction.

L_{it}.[21] To be consistent with the estimation of relative demand shocks, I use annual data. From 1973–2019, annual productivity growth volatility was 1.2% in both the US and in the aggregate of G10 economies, with an international correlation of 0.50. This pins down the volatility of the idiosyncratic and global productivity shocks σ_{Ia} and σ_G. For the persistence of productivity ρ_a, I target the persistence of HP-detrended annual US productivity of 0.47.[22]

I set relative risk aversion $\gamma = 2$, a standard value in the literature. The disutility of labor is also constant elasticity: $v(L) = \chi_0 L^{1+1/\chi_1}/(1 + 1/\chi_1)$. The Frisch elasticity of labor supply χ_1 targets the volatility of hours worked to output, and the weight of labor in utility χ_0 targets working hours of 20% of total available time.[23] The labor share α, capital depreciation rate δ, and time preference β target standard moments from the business cycle literature. I normalize the initial value of each country's portfolio $\bar{B}_i$ so that both countries have equal Pareto weights in the risk-sharing condition: $\mu_i \mathcal{E}_{ij}(s^0) = \mu_j$ in equation (40).

5 Quantitative Model: Findings

I solve the model with the parameters in Table 1. In Table 3, I compare the model's exchange rate moments (column 2) to the data (column 1). For each puzzle, I focus on the median statistic from 100 simulations of the model of the same length as the data. For

[21]The capital stock has low volatility at short-run frequencies and is not particularly correlated with business cycles, so makes little difference if omitted. I use employment instead of hours worked because time series on hours worked are not generally internationally comparable.

[22]As an added check, Appendix Figure 13 verifies that this measure of US TFP is largely consistent with the series calculated by Fernald (2014).

[23]Although the Frisch elasticity is somewhat larger than microeconometric estimates of around 1.0, it is conservative: with a smaller Frisch elasticity (and a mechanism that amplifies fluctuations in hours worked to reach its empirical level), pro-cyclical fluctuations in hours worked would more strongly dampen the cyclicality of marginal utility in equation (43), which would make the Backus-Smith puzzle easier to resolve.

Table 3: US Real Exchange Rate Puzzles

	Data	Model	Variants of the Model		
			Standard	No fin. fric.	Fin. autarky
	(1)	(2)	(3)	(4)	(5)
Cyclicality: Backus-Smith puzzle					
$\text{corr}\big[\Delta \log(C_j/C_i),\ \Delta \log(\mathcal{Q}_{ij})\big]$	0.19	0.18	-0.87	-0.07	0.79
		[0.06, 0.29]	[-0.85, -0.90]	[-0.17, 0.02]	[0.75, 0.84]
Volatility of exchange rate to GDP					
$\sigma[\Delta \log \mathcal{Q}_{ij}]/\sigma[\Delta \log Y_i]$	3.9	2.5	0.3	1.9	4.1
		[1.9, 3.3]	[0.2, 0.4]	[1.5, 2.5]	[3.1, 5.7]
Persistence of RER fluctuations *PPP puzzle (Rogoff, 1996)*					
half life of RER fluctuations	17 qtrs	13 qtrs	–	14 qtrs	13 qtrs
		[5, 27]		[6, 33]	[6, 23]
ζ		21	∞	∞	–
$Z_{ij} - Z_{ii},\ j \neq i$		0.11	0	0	∞
σ_ω (%)		8.9	0	8.9	8.9

Notes: data 1973–2019, half life is the first t such that $\text{corr}(y_t, y_{t-1})^t \leq 0.5$, main figures are the 50th percentile from 100 simulations of same length as data, brackets indicate 5th and 95th percentiles.

the Backus-Smith puzzle, my model has a modest positive correlation between j's relative consumption and j's real exchange rate appreciations: $\text{corr}\big[\Delta \log(C_j/C_i),\ \Delta \log(\mathcal{Q}_{ij})\big]$ is 0.18, close to its correlation in the data of 0.19.[24] I find an annual volatility of exchange rates 2.5 that of output, close to but somewhat below the value in the data. For the purchasing power parity puzzle, the half-life of the impulse response to a shock to ω_i is 13 quarters, which is close to the 17 quarters in the data. Hence, the model reproduces three major exchange rate puzzles.

Conversely, the canonical international real business cycle model, which has neither financial frictions nor relative demand shocks (column 3), produces the Backus-Smith puzzle with a near-perfect correlation between relative consumption and exchange rate depreciation. It also produces almost no volatility in exchange rates, as productivity shocks have a relatively muted impact on international prices. Adding relative demand shocks without financial frictions (column 4) makes exchange rates volatile, but is unable to resolve the Backus-Smith puzzle. At the other extreme, a model with full financial autarky, so that $Z_{ij} = \infty$ for any $j \neq i$, makes the correlation between relative consumption and the real exchange rate much too strong.

In Table 4, I report business cycle correlations. Domestic business cycle moments between output, consumption, employment, and investment match the data fairly closely. There are two reasons for this: Backus, Kehoe and Kydland (1992) show that the international real business cycle model driven by productivity shocks fits the business cycle moments well. The addition of the relative demand shock does not impair its

[24]This value is similar to Corsetti, Dedola and Leduc (2008), who calculate a correlation of 0.25.

Table 4: US Business Cycle Moments

Variable		Volatility relative to GDP		Correlation with GDP		International correlation	
		Data	Model	Data	Model	Data	Model
Y	GDP	3.0%	3.4%	1.00	1.00	0.68	0.49
			[2.9, 4.0]				[0.29, 0.64]
C	Consumption	0.81	0.61	0.87	0.99	0.53	0.60
			[0.60, 0.62]		[0.99, 1.00]		[0.42, 0.72]
L	Employment	0.78	0.66	0.82	0.97	0.73	0.35
			[0.62, 0.70]		[0.96, 0.98]		[0.12, 0.52]
I^K	Investment	2.71	2.25	0.94	0.94	0.59	0.42
			[2.08, 2.44]		[0.91, 0.96]		[0.20, 0.57]
NX	Net exports	3.48	1.87	−0.37	0.17		
			[1.50, 2.36]		[−0.07, 0.37]		

Notes: data 1973–2019, HP filtered with $\lambda = 1600$, see footnote 17 for construction of the foreign data, main figures are the 50th percentile from 100 simulations of same length as data, brackets indicate 5th and 95th percentiles.

Table 5: International Risk Sharing

		Variants of the Model		
	Model	Standard	No fin. fric.	Fin. autarky
	(1)	(2)	(3)	(4)
Marginal utility per dollar				
$\mathrm{corr}\big(\Delta\log\big(\frac{u'_j}{\mathcal{E}_{ij}P_j^F}\big),\ \Delta\log\big(\frac{u'_i}{P_i^F}\big)\big)$	0.96	1.00	1.00	−0.00
	[0.93, 0.95]			[−0.13, 0.12]
Marginal utility				
$\mathrm{corr}\big(\Delta\log u'_j,\ \Delta\log u'_i\big)$	−0.23	0.89	−0.63	0.22
	[−0.35, 0.06]	[0.84, 0.92]	[−0.71, −0.55]	[0.11, 0.30]
ζ	21	∞	∞	–
$Z_{ij} - Z_{ii},\ j \neq i$	0.11	0	0	∞
σ_ω (%)	8.9	0	8.9	8.9

Notes: main figures are the 50th percentile from 100 simulations of same length as data, brackets indicate 5th and 95th percentiles.

performance because relative demand shocks are qualitatively similar to productivity shocks: a positive shock $\omega_{it} > 0$ increases i's output, consumption (only with international financial frictions), labor, and investment. The crucial difference is that it causes i's exchange rate to appreciate, instead of depreciate.[25]

Finally, in Table 5, the model suggests that international risk sharing remains robust despite international financial frictions: marginal utility *per dollar* is highly correlated across countries. However, this does not extend to marginal utility itself, which is negatively correlated due to fluctuations in the real exchange rate (see eqn. 40). These results suggest that small deviations from full international risk sharing are sufficient to resolve the Backus-Smith puzzle. Under financial autarky (column 4), the co-movement in the marginal utility per dollar across countries is low. This finding is the opposite to Cole and Obstfeld (1991), who show that with only productivity shocks and no relative demand shocks, movements in the real exchange rate effectively implement international risk sharing even in financial autarky.

5.1 Reconstructing Shocks with a Kalman Filter

In this section, I reconstruct through the lens of the model the forces that drive the data. The Kalman filter uses the decision rules of the model to back out the most likely shock that drives the data. I apply the Kalman filter to data on three time series: $\left[c_{it}, c_{jt}, q_{jt}\right]$, which are US log consumption, foreign (G10 excluding the US) log consumption, and the US real exchange rate, and use it to back out three time series of shocks $\left[\varepsilon^a_{it}, \varepsilon^a_{jt}, \varepsilon^\omega_t\right]$, which are US productivity shocks, foreign productivity shocks, and US relative demand shocks.[26]

I start by verifying that given the shocks backed out via the Kalman filter, the model matches the data well (see Appendix Figure 15). Then Figure 6 decomposes time series into their underlying drivers. Consistent with the international real business cycle literature, US productivity shocks (in blue) drive most fluctuations in US GDP (top left) and US consumption (bottom left), while foreign productivity shocks (in orange) drive most fluctuations in foreign GDP (top right). However, domestic productivity shocks drive little of real exchange rate movements, which the Kalman filter overwhelmingly attributes to relative demand shocks (in green, bottom right panel). These results are consistent with the impulse response functions in Appendix Figure 14, which show that domestic productivity shocks have larger effects on macroeconomic quantities, but relative demand shocks have significantly larger effects on real exchange rates. The time series of the shocks are plotted in Appendix Figure 16. Consistent with the analysis in Figure 6, the backed-out series of US productivity shocks track US GDP and consumption closely, foreign productivity shocks track foreign GDP and consumption closely, and relative demand shocks track the US real exchange rate closely.

[25]The international correlations inherit the puzzles of the canonical international real business cycle model described in Backus, Kehoe and Kydland (1993): consumption is more internationally correlated than output, employment is negatively correlated across countries, and net exports are not countercyclical. A large literature explores mechanisms to resolve each of these puzzles. As an example for net exports, see Drozd and Nosal (2012).

[26]As the final goods input weights η_{ij} and η_{ij} are normalized so that only the ratios of ω_j matter (and not their levels), a separate time series for foreign relative demand cannot be identified.

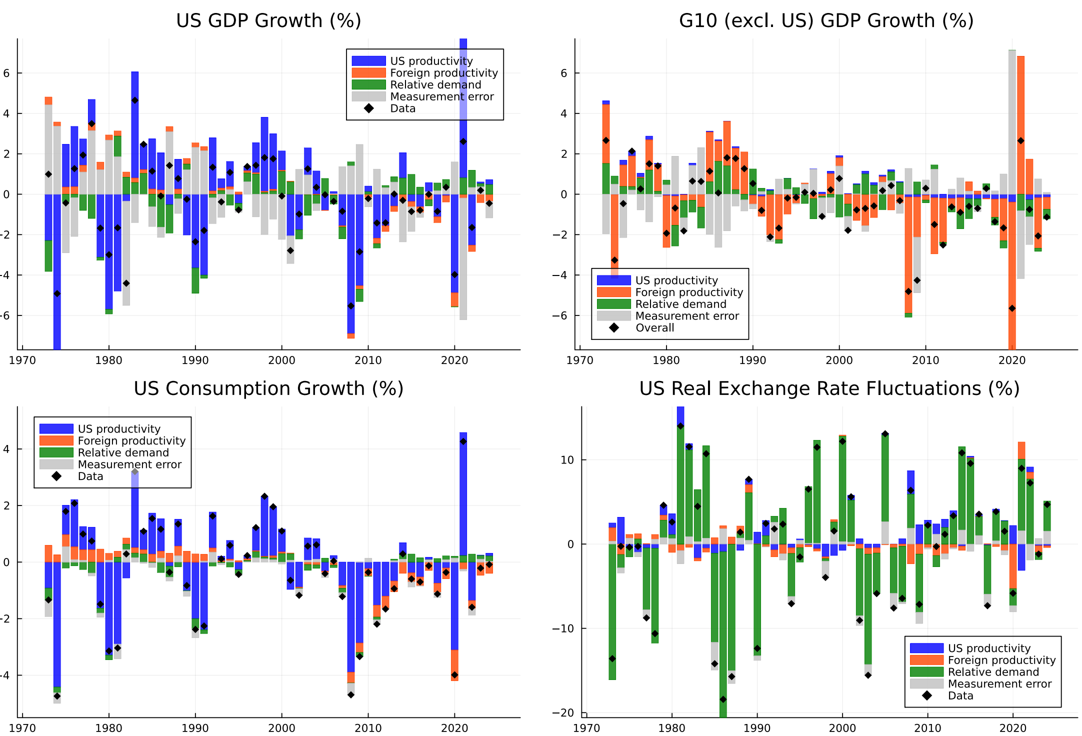

Figure 6: Kalman Filter: Shock Decomposition

Notes: Decomposition of shocks driving US GDP growth (*top left*), foreign GDP growth (*top right*), US consumption growth (*bottom left*), and US real exchange rate $1/\mathcal{Q}_{ij}$ fluctuations (*bottom right*).

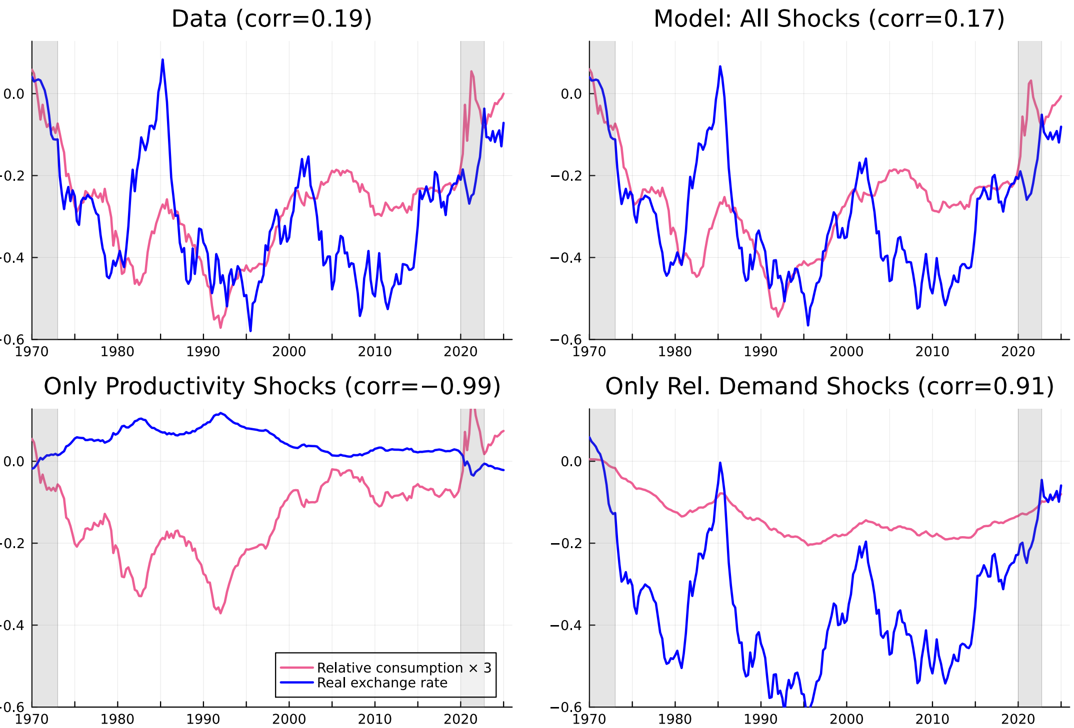

Figure 7: Kalman Filter: Decomposition of Relative Consumption and the US Real Exchange Rate

Notes: relative consumption is $\log(C_i/C_j)$ (higher indicates higher US consumption relative to aggregate of G10 countries) and real exchange rate is $1/\mathcal{Q}_{ij}$ (higher indicates US appreciation). Bottom-left panel: counterfactual time series with relative demand shocks turned off: $\omega_i \equiv 0$. Bottom-right panel: counterfactual time series with productivity shocks turned off: $a_i \equiv 0$.

Next, I decompose the contribution of the two types of shock to illustrate how the mechanism works. In Figure 7, the top left panel plots the US real exchange rate $1/q_{aj}$ in blue (higher indicates US appreciation) against US consumption relative to foreign consumption in pink. The top right panel shows that the model with all shocks reproduces the data well. The bottom left panel plots a counterfactual series with only productivity shocks: it greatly reduces the volatility of the exchange rate, and it produces a very negative -0.99 correlation between relative consumption and exchange rate appreciations, following the logic of Section 3.6 that a productivity shock increases the supply of a country's intermediate goods and lowers their price. This is in line with the international business cycle literature, which struggles to match exchange rate patterns with productivity shocks. The bottom right panel plots a counterfactual series with only relative demand shocks: now the correlation between relative consumption and exchange rate appreciations is too strong, and the fit of the relative consumption series is poor. With both shocks, the model produces the right correlation between relative consumption and exchange rates, which is in between these two extremes.

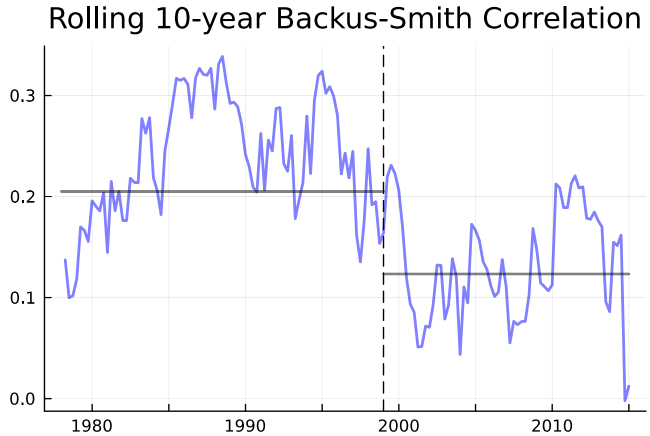

Figure 8: Rolling 10-year Backus-Smith Correlation

Notes: $\text{corr}(\Delta\log(C_i/C_j), \Delta\log(1/\mathcal{Q}_{ij}))$ in the data for US vs an aggregate of G10 currencies. The x-axis plots the centre of the 10-year rolling window.

Table 6: Subsample Analysis Quantification

Parameter		Value	Moment	Target
		Early period (1973–1998)		
Z_{ij}	foreign financial friction	1.25	$\text{E}[HB_{it}]$	91%
$\bar{\eta}_{ij}$	expenditure weight: imports	8.3%	$1 - \text{E}[P^X_{it}X_{iit}/P^F_{it}(C_{it} + I^K_{it})]$	8.3%
		Recent period (1999–2019)		
Z_{ij}	foreign financial friction	1.15	$\text{E}[HB_{it}]$	63%
$\bar{\eta}_{ij}$	expenditure weight: imports	12.1%	expenditure weight: imports	12.1%

5.2 Subsample Analysis

The key testable prediction of the model is that the extent of international risk sharing determines the strength of the Backus-Smith correlation. Figure 4 shows that US holdings of foreign securities have increased over time, especially through the 1990s and 2000s, so the US aggregate portfolio return has more exposure to the returns on foreign securities. In the model, this diminishes the portfolio return effect in the international risk sharing equation (42), so the correlation between movements in a country's relative consumption and appreciation in its exchange rate should decline.

Figure 8 shows that the prediction of a decline in the Backus-Smith correlation is borne out in the data. The decline is most significant through the 1990s and 2000s, during which most movement in the US foreign portfolio share occurs.

I decompose the change in this correlation into the component driven by globalization and the component driven by a change in the underlying shock process. To do so, I re-quantify the model for the 1973–1998 period and the 1999–2019 period, focusing on the change in trade and financial openness. The changed parameters are in Table 6. To construct a counterfactual series for the Backus-Smith correlation in the absence

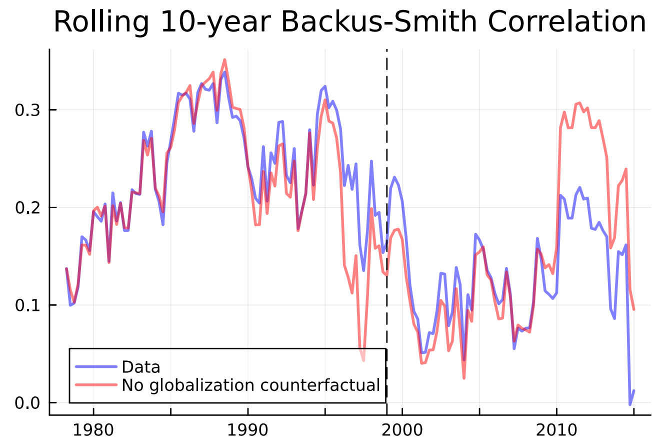

Figure 9: Rolling 10-year Backus-Smith Correlation: Data vs Counterfactual

Notes: $\text{corr}(\Delta \log(C_i/C_j), \Delta \log(1/\mathcal{Q}_{ij}))$ for US vs an aggregate of G10 currencies. The counterfactual of no globalization uses the early period's trade and financial openness parameters in Table 6 to look at the shocks in the data. The x-axis plots the centre of the 10-year rolling window.

of changes in openness, I first construct a time series of actual shocks by combining the 1973–1998 time series of shocks recovered by applying the Kalman filter with the early period parameters with the 1999–2019 time series of shocks recovered by applying the Kalman filter with the recent period parameters; I then apply the Kalman filter with the early period parameters over the entire time series to obtain the no-openness counterfactual.

Figure 9 shows the counterfactual Backus-Smith correlation in the absence of financial globalization in red. In the recent period, it is notably higher than the data in blue, although still somewhat lower than in the early period (the Kalman filter attributes the remainder of the decline to changes in the shocks).

6 Alternative Mechanisms

Itskhoki and Mukhin (2021) and Kekre and Lenel (2024a) construct models where international financial markets are closed to investors, so that US investors cannot buy foreign assets, and vice versa. Instead, the only asset available to households are domestic nominal risk-free bonds. International asset trade is limited to specialized intermediaries, who trade the nominal risk-free bonds. They show that in such settings, shocks to Z_{ij} (which may be interpreted as foreign intermediation costs or as shocks to the behavior of noise traders) and to discount rates β can produce exchange rate dynamics seen in the data.

I put these shocks into my model by allowing $Z_{ij}(s^t)$ $(j \neq i)$ and $\beta_i(s^t)$ follow an AR(1) process in the representative household's problem. I calibrate their persistence to match the annualized persistence of relative demand shocks ρ_ω, which is representative of the values they choose. I calibrate the size of their shocks as follows: for foreign intermediation shocks, I follow Itskhoki and Mukhin (2021) by targeting a Backus-Smith correlation of

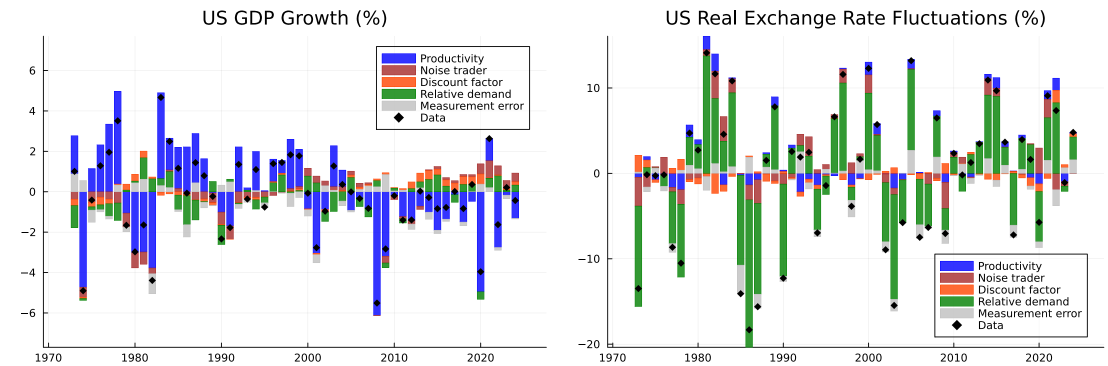

Figure 10: Kalman Filter with Additional Shocks: Shock Decomposition

Notes: Decomposition of shocks driving US GDP growth (*left*) and US real exchange rate fluctuations $1/\mathcal{Q}_{ij}$ (*right*, higher indicates US appreciation), with the addition of noise trader and discount factor shocks.

-0.40 in a model that consists only of productivity and foreign intermediation shocks, which obtains a 5% annualized volatility of shocks to $Z_{ij}(s^t)$. I set the annualized volatility of discount factor shocks to 0.4%, in line with Kekre and Lenel (2024a).

I re-run the Kalman filter to back out these new shocks, adding US and foreign GDP (Y) and capital investment (I^K) as observable series. The shock decomposition for US GDP and real exchange rate fluctuations are shown in Figure 10. The results look very similar to the previous shock decomposition in Figure 6: productivity shocks drive most fluctuations in GDP growth, while relative demand shocks drive most fluctuations in the real exchange rate.

Hence, these empirical results suggest that in a model with relative demand shocks, there is little room for foreign intermediation cost/noise trader shocks and discount factor shocks to explain exchange rate movements.[27] The reason for this is because when investors can trade more than just one-period nominal risk-free bonds, as they must to satisfy the documented portfolio facts, and under the empirically relevant level of trade openness, these alternative shocks are incapable of generating large exchange rate movements without generating similarly-large movements in consumption.

I plot the initial $t = 0$ impulse responses of the exchange rate and relative consumption to each of these shocks in Figure 11. Noise trader (row 1) and discount factor shocks (row 2) are only capable of generating large exchange rate movements compared to consumption when trade openness (left column, import composition of consumption $1 - \eta_{ii}^F$) is far below its value in the data, e.g. in the *autarky limit* where $1 - \eta_{ii}^F \to 0$. In contrast, relative demand shocks (row 3) can generate large exchange rate movements compared to consumption. Variation in the level of home portfolio bias (right column) has little effect on the relative size of exchange rate movements and consumption, and instead scales up both impulse responses roughly in proportion.

[27] The results also show that productivity shocks cannot drive most movements in the real exchange rate, contrary to Corsetti, Dedola and Leduc (2008).

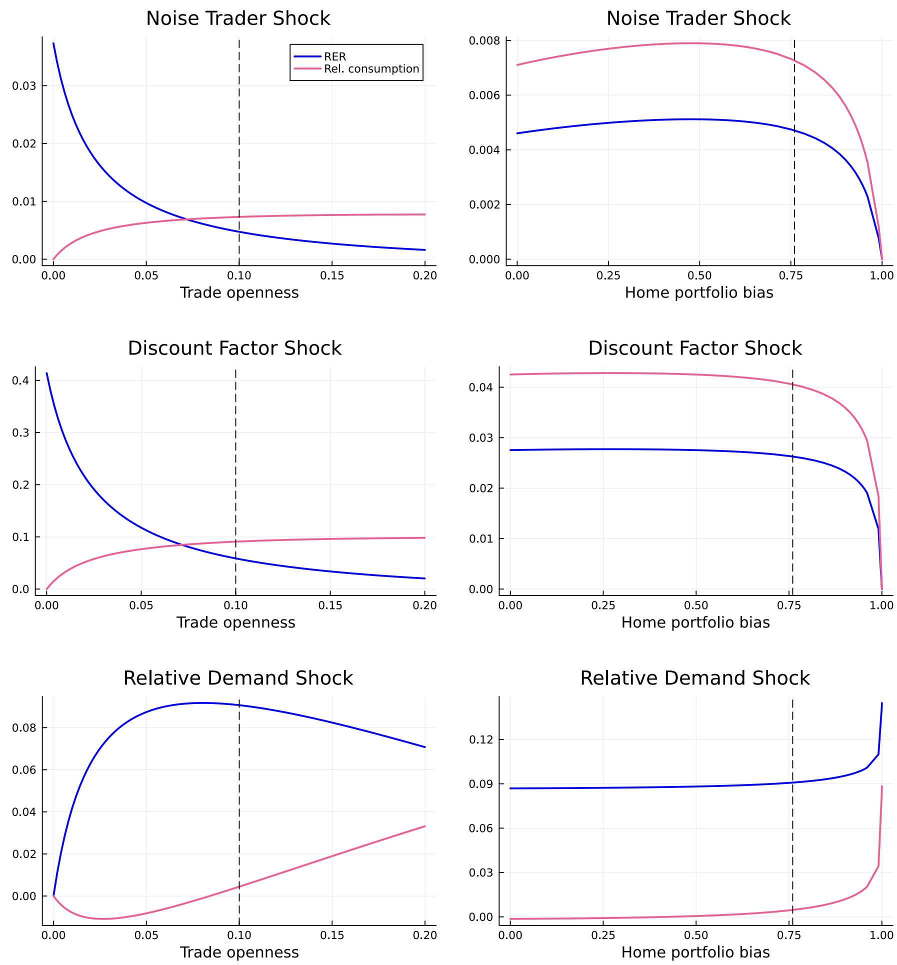

Figure 11: Alternative Shocks: Impulse Response of Real Exchange Rate and Relative Consumption

Notes: left column: variation in trade openness $1 - \eta_{ii}^F$. Right column: variation in home portfolio bias from varying Z_{ij} $(j \neq i)$. First row: a 10ppt increase in Z_{ij} $(j \neq i)$. Second row: a 20% increase in β_i. Third row: a 20% increase in ω_i. I turn off the investment channel in this exercise.

7 Conclusion

A central question in international macroeconomics is the extent to which international financial markets allow countries to mitigate aggregate shocks, with the Backus-Smith puzzle—increases in a country's relative consumption are correlated with its exchange rate appreciating—being the key piece of evidence in favor of limited risk sharing. I show that in a model with international financial frictions, with foreign portfolio holdings as they are in the data, relative demand shocks resolve the Backus-Smith puzzle. The mechanism operates via changes in wealth: when demand for American goods increases, American firms become more profitable and American securities pay investors higher returns. Under home portfolio bias, American households own most American securities, so they become relatively wealthier and consume more. Hence, American households increase their relative consumption following a US real exchange rate appreciation, resolving the Backus-Smith puzzle.

References

Acharya, Sushant, Edouard Challe, and Louphou Coulibaly, "International Risk Sharing and the Transmission of Shocks Redux," Working Paper January 2025.

Alvarez, Fernando, Andrew Atkeson, and Patrick J. Kehoe, "Time-Varying Risk, Interest Rates, and Exchange Rates in General Equilibrium," *The Review of Economic Studies*, 07 2009, *76* (3), 851–878.

_ , _ , and Patrick J. Kehoe, "Money, Interest Rates, and Exchange Rates with Endogenously Segmented Markets," *Journal of Political Economy*, 2002, *110* (1), 73–112.

Arkolakis, Costas, Arnaud Costinot, and Andrés Rodríguez-Clare, "New Trade Models, Same Old Gains?," *American Economic Review*, February 2012, *102* (1), 94–130.

Armington, Paul S., "A Theory of Demand for Products Distinguished by Place of Production," *Staff Papers (International Monetary Fund)*, 1969, *16* (1), 159–178.

Arrow, K. J., "The Role of Securities in the Optimal Allocation of Risk-bearing," *The Review of Economic Studies*, 1964, *31* (2), 91–96.

Bacchetta, Philippe and Eric van Wincoop, "Infrequent Portfolio Decisions: A Solution to the Forward Discount Puzzle," *American Economic Review*, June 2010, *100* (3), 870–904.

Backus, David K. and Gregor W. Smith, "Consumption and real exchange rates in dynamic economies with non-traded goods," *Journal of International Economics*, 1993, *35* (3), 297–316.

_ , Patrick J. Kehoe, and Finn E. Kydland, "International Real Business Cycles," *Journal of Political Economy*, 1992, *100* (4), 745–775.

Backus, David, Patrick J Kehoe, and Finn E Kydland, "International Business Cycles: Theory and Evidence," Working Paper 4493, National Bureau of Economic Research October 1993.

Balassa, Bela, "The Purchasing-Power Parity Doctrine: A Reappraisal," *Journal of Political Economy*, 1964, *72* (6), 584–596.

Baxter, Marianne and Mario J. Crucini, "Business Cycles and the Asset Structure of Foreign Trade," *International Economic Review*, 1995, *36* (4), 821–854.

_ and Urban J. Jermann, "The International Diversification Puzzle Is Worse Than You Think," *The American Economic Review*, 1997, *87* (1), 170–180.

Bureau of Economic Analysis, "Table 2.4.5. Personal Consumption Expenditures by Type of Product," 2025.

Caliendo, Lorenzo and Fernando Parro, "Estimates of the Trade and Welfare Effects of NAFTA," *The Review of Economic Studies*, 2015, *82* (1 (290)), 1–44.
Camanho, Nelson, Harald Hau, and Hélène Rey, "Global Portfolio Rebalancing and Exchange Rates," *The Review of Financial Studies*, 04 2022, *35* (11), 5228–5274.
Chari, V. V, Patrick J. Kehoe, and Ellen R. McGrattan, "Can Sticky Price Models Generate Volatile and Persistent Real Exchange Rates?," *The Review of Economic Studies*, 07 2002, *69* (3), 533–563.
Coeurdacier, Nicolas and Hélène Rey, "Home Bias in Open Economy Financial Macroeconomics," *Journal of Economic Literature*, March 2013, *51* (1), 63–115.
_ **and Pierre-Olivier Gourinchas**, "When bonds matter: Home bias in goods and assets," *Journal of Monetary Economics*, 2016, *82*, 119–137.
Colacito, Ric, Max Croce, Steven Ho, and Philip Howard, "BKK the EZ Way: International Long-Run Growth News and Capital Flows," *American Economic Review*, November 2018, *108* (11), 3416–49.
Colacito, Riccardo and Mariano M. Croce, "Risks for the Long Run and the Real Exchange Rate," *Journal of Political Economy*, 2011, *119* (1), 153–181.
_ **and** _ , "International Asset Pricing with Recursive Preferences," *The Journal of Finance*, 2013, *68* (6), 2651–2686.
Cole, Harold L. and Maurice Obstfeld, "Commodity trade and international risk sharing: How much do financial markets matter?," *Journal of Monetary Economics*, 1991, *28* (1), 3–24.
Coppola, Antonio, Matteo Maggiori, Brent Neiman, and Jesse Schreger, "Redrawing the Map of Global Capital Flows: The Role of Cross-Border Financing and Tax Havens*," *The Quarterly Journal of Economics*, 05 2021, *136* (3), 1499–1556.
Corsetti, Giancarlo, Luca Dedola, and Sylvain Leduc, "International Risk Sharing and the Transmission of Productivity Shocks," *The Review of Economic Studies*, 04 2008, *75* (2), 443–473.
Debreu, Gerard, *Theory of Value*, New Haven, Connecticut: Yale University Press, 1959.
Dekle, Robert, Jonathan Eaton, and Samuel Kortum, "Global Rebalancing with Gravity: Measuring the Burden of Adjustment," *IMF Staff Papers*, 2007, *55* (3), 511–540.
Diamond, Peter A., "The Role of a Stock Market in a General Equilibrium Model with Technological Uncertainty," *The American Economic Review*, 1967, *57* (4), 759–776.
Drozd, Lukasz A. and Jaromir B. Nosal, "Understanding International Prices: Customers as Capital," *American Economic Review*, February 2012, *102* (1), 364–95.
Engel, Charles, "Accounting for U.S. Real Exchange Rate Changes," *Journal of Political Economy*, 1999, *107* (3), 507–538.
_ **and Kenneth D. West**, "Exchange Rates and Fundamentals," *Journal of Political Economy*, 2005, *113* (3), 485–517.
Feenstra, Robert C., Philip Luck, Maurice Obstfeld, and Katheryn N. Russ, "In Search of the Armington Elasticity," *The Review of Economics and Statistics*, 03 2018, *100* (1), 135–150.
Fernald, John, "A Quarterly, Utilization-Adjusted Series on Total Factor Productivity," Working Paper 2012-19, Federal Reserve Bank of San Francisco April 2014.
French, Kenneth R. and James M. Poterba, "Investor Diversification and International Equity Markets," *The American Economic Review*, 1991, *81* (2), 222–226.
Fukui, Masao, Emi Nakamura, and Jón Steinsson, "The Macroeconomic Consequences of Exchange Rate Depreciations," Working Paper 31279, National Bureau of Economic Research May 2023.
Gabaix, Xavier and Matteo Maggiori, "International Liquidity and Exchange Rate Dynamics," *The Quarterly Journal of Economics*, 03 2015, *130* (3), 1369–1420.
Gopinath, Gita and Oleg Itskhoki, "Frequency of Price Adjustment and Pass-Through*,"

The Quarterly Journal of Economics, 05 2010, *125* (2), 675–727.

Greenwood, Jeremy, Zvi Hercowitz, and Gregory W. Huffman, "Investment, Capacity Utilization, and the Real Business Cycle," *The American Economic Review*, 1988, *78* (3), 402–417.

Guo, Xing, Pablo Ottonello, and Diego J. Perez, "Monetary Policy and Redistribution in Open Economies," *Journal of Political Economy Macroeconomics*, 2023, *1* (1), 191–241.

Heathcote, Jonathan and Fabrizio Perri, "Financial autarky and international business cycles," *Journal of Monetary Economics*, 2002, *49* (3), 601–627.

_ and _ , "The International Diversification Puzzle Is Not as Bad as You Think," *Journal of Political Economy*, 2013, *121* (6), 1108–1159.

International Monetary Fund, "International Financial Statistics (IFS) Database," 2025.

Itskhoki, Oleg and Dmitry Mukhin, "Exchange Rate Disconnect in General Equilibrium," *Journal of Political Economy*, 2021, *129* (8), 2183–2232.

Jiang, Zhengyang, Arvind Krishnamurthy, and Hanno Lustig, "Implications of Asset Market Data for Equilibrium Models of Exchange Rates," Working Paper 31851, National Bureau of Economic Research November 2023.

_ , _ , _ , and Jialu Sun, "Convenience Yields and Exchange Rate Puzzles," Working Paper 32092, National Bureau of Economic Research January 2024.

_ , Robert J. Richmond, and Tony Zhang, "A Portfolio Approach to Global Imbalances," *The Journal of Finance*, 2024, *79* (3), 2025–2076.

_ , _ , and _ , "Understanding the strength of the dollar," *Journal of Financial Economics*, 2025, *168*, 104052.

Kehoe, Patrick J. and Fabrizio Perri, "International Business Cycles with Endogenous Incomplete Markets," *Econometrica*, 2002, *70* (3), 907–928.

Kekre, Rohan and Moritz Lenel, "Exchange Rates, Natural Rates, and the Price of Risk," Working Paper 32976, National Bureau of Economic Research September 2024.

_ and _ , "The Flight to Safety and International Risk Sharing," *American Economic Review*, June 2024, *114* (6), 1650–91.

Kleinman, Benny, Ernest Liu, Stephen J Redding, and Motohiro Yogo, "Neoclassical Growth in an Interdependent World," Working Paper 31951, National Bureau of Economic Research December 2023.

Koijen, Ralph S. J and Motohiro Yogo, "Exchange Rates and Asset Prices in a Global Demand System," Working Paper 27342, National Bureau of Economic Research June 2020.

Kollmann, Robert, "Incomplete asset markets and the cross-country consumption correlation puzzle," *Journal of Economic Dynamics and Control*, 1996, *20* (5), 945–961.

Lewis, Karen K., "Trying to Explain Home Bias in Equities and Consumption," *Journal of Economic Literature*, June 1999, *37* (2), 571–608.

Lucas, Robert E., "Equilibrium in a Pure Currency Economy," *Economic Inquiry*, 1980, *18* (2), 203–220.

_ , "Interest rates and currency prices in a two-country world," *Journal of Monetary Economics*, 1982, *10* (3), 335–359.

Lustig, Hanno and Adrien Verdelhan, "Does Incomplete Spanning in International Financial Markets Help to Explain Exchange Rates?," *American Economic Review*, June 2019, *109* (6), 2208–44.

Marin, Emile A. and Sanjay R. Singh, "Incomplete Markets and Exchange Rates," Working Paper Series 2025-11, Federal Reserve Bank of San Francisco Jul 2025.

McFadden, Daniel, "Conditional logit analysis of qualitative choice behavior," Technical Report 105-142, Frontiers in Econometrics, New York 1974.

Meese, Richard A. and Kenneth Rogoff, "Empirical exchange rate models of the seventies: Do they fit out of sample?," *Journal of International Economics*, 1983, *14* (1), 3–24.

Obstfeld, Maurice and Kenneth Rogoff, "The Six Major Puzzles in International Macroeconomics: Is There a Common Cause?," *NBER Macroeconomics Annual*, 2000, *15*, 339–390.

Oviedo, P. Marcelo and Rajesh Singh, "Investment composition and international business cycles," *Journal of International Economics*, 2013, *89* (1), 79–95.

Pavlova, Anna and Roberto Rigobon, "Asset Prices and Exchange Rates," *The Review of Financial Studies*, 01 2007, *20* (4), 1139–1180.

Rogoff, Kenneth, "The Purchasing Power Parity Puzzle," *Journal of Economic Literature*, 1996, *34* (2), 647–668.

Samuelson, Paul A., "Theoretical Notes on Trade Problems," *The Review of Economics and Statistics*, 1964, *46* (2), 145–154.

Verdelhan, Adrien, "A Habit-Based Explanation of the Exchange Rate Risk Premium," *The Journal of Finance*, 2010, *65* (1), 123–146.

World Federation of Exchanges, "Market capitalization of listed domestic companies," Published by World Bank 2025.

A Derivations

A.1 Proof of Proposition 1 (Portfolio Demand)

Restatement of proposition: household ι's problem is to choose paths of consumption of goods $C_i^g(\iota, s^t)$, labor $L_i(\iota, s^t)$, and asset portfolio $D_{ij}(\iota, s^\tau \mid s^t)$ to maximize its expected utility

$$\sum_t \sum_{s^t} \beta^t \pi(s^t) u\big(C_i^g(\iota, s^t) - v(L_i(\iota, s^t))\big), \tag{45}$$

subject to cash-in-advance constraint

$$P_i^F(s^t) C_i^g(\iota, s^t) = \sum_j \left[\mathbf{1}_{D_{ij}(\iota, s^t) > 0} \frac{\mathcal{E}_{ij}(s^t) D_{ij}(\iota, s^t)}{Z_{ij}(\iota)} + \mathbf{1}_{D_{ij}(\iota, s^t) < 0} \mathcal{E}_{ij}(s^t) D_{ij}(\iota, s^t) \right] + \frac{W_i(s^t) L_i(\iota, s^t)}{Z_i^L}. \tag{46}$$

where $D_{ij}(\iota, s^t) = D_{ij}(\iota, s^t \mid s^t)$, and a sequence of budget constraints

$$\sum_{\tau \geq t} \sum_{s^\tau} Q_j(s^\tau \mid s^t) D_{ij}(\iota, s^\tau \mid s^t) \leq \sum_{\tau \geq t} \sum_{s^\tau} Q_j(s^\tau \mid s^t) D_{ij}(\iota, s^\tau \mid s^{t-1}). \tag{47}$$

$$\sum_t \sum_{s^t} \sum_j \mathcal{E}_{ij}(s^0) Q_j(s^t \mid s^0) D_{ij}(\iota, s^t \mid s^0) \leq \bar{B}_i(\iota) \tag{48}$$

and no-Ponzi scheme constraint $\lim_{\tau \to \infty} D_{ij}(\iota, s^\tau \mid s^t) \geq 0$. Assume that $Z_{ij}(\iota)$ is distributed according to:

$$\frac{1}{Z_{ij}(\iota)} \sim \text{Fréchet}\left(\frac{\kappa_0}{Z_{ij}}, \frac{1}{\zeta} \right),$$

where the constant $\kappa_0 = (\frac{\Gamma(\zeta/(\zeta-1))}{\Gamma((\zeta-1/\gamma)/(\zeta-1))})^\gamma$, and that the initial wealth of households are such that all households in country i have the same marginal utility of wealth μ_i.

Then, country i's aggregate portfolio share in country j's dividend strips is

$$d_{ij}^{sh}(s^t) = \left(\frac{R_{ij}^D(s^t)/Z_{ij}}{R_i^H(s^t)} \right)^\zeta, \tag{49}$$

where i's aggregate portfolio return is

$$R_i^H(s^t) = \left(\sum_j \left[R_{ij}^D(s^t)/Z_{ij} \right]^{\zeta - 1} \right)^{\frac{1}{\zeta - 1}}. \tag{50}$$

Step 1: Reformulate household problem in real terms. For ease of exposition, I rewrite the problem of household ι of country i in real terms, so that all quantities are expressed in terms of country i's final goods. In this section, let lowercase variables denote real variables:

$$d_{ij}(\iota, s^\tau \mid s^t) = \frac{\mathcal{E}_{ij}(s^\tau) D_{ij}(\iota, s^\tau \mid s^t)}{P_i^F(s^\tau)} \tag{51}$$

$$q_{ij}(s^t) = \frac{\mathcal{E}_{ij}(s^0) Q_j(s^t)}{\mathcal{E}_{ij}(s^t)} \frac{P_i^F(s^t)}{P_i^F(s^0)} = \frac{1}{R_{ij}^D} \frac{P_i^F(s^t)}{P_i^F(s^0)} \tag{52}$$

$$w_i(s^t) = \frac{W_i(s^t)}{P_i^F(s^t)} \tag{53}$$

Using this notation, the household's cash-in-advance constraint (46) becomes

$$C_i^g(\iota, s^t) = \sum_j \frac{d_{ij}(\iota, s^t)}{Z_{ij}(\iota)} + \frac{w_i(s^t) L_i(\iota, s^t)}{Z_i^L}.$$

Before proceeding with the budget constraints, I follow the usual steps of recursively substituting the time-t budget constraint (47) into the time-0 budget constraint (48) to show that the time-0 constraint suffices to capture the entire sequence of constraints:

$$\begin{aligned}
\bar{B}_i(\iota) &\geq \sum_j \sum_\tau \sum_{s^\tau} \mathcal{E}_{ij}(s^0) Q_j(s^\tau \mid s^0) D_{ij}(\iota, s^\tau \mid s^0) \\
&= \sum_j \mathcal{E}_{ij}(s^0) \Bigg[Q_j(s^0 \mid s^0) D_{ij}(\iota, s^0 \mid s^0) + \sum_{\tau \geq 1} \sum_{s^\tau} Q_j(s^\tau \mid s^0) D_{ij}(\iota, s^\tau \mid s^0) \Bigg] \\
&= \sum_j \mathcal{E}_{ij}(s^0) \Bigg[Q_j(s^0 \mid s^0) D_{ij}(\iota, s^0 \mid s^0) + Q_j(s^1 \mid s^0) \sum_{\tau \geq 1} \sum_{s^\tau} Q_j(s^\tau \mid s^1) D_{ij}(\iota, s^\tau \mid s^0) \Bigg] \\
&\geq \sum_j \mathcal{E}_{ij}(s^0) \Bigg[Q_j(s^0 \mid s^0) D_{ij}(\iota, s^0 \mid s^0) + Q_j(s^1 \mid s^0) \sum_{\tau \geq 1} \sum_{s^\tau} Q_j(s^\tau \mid s^1) D_{ij}(\iota, s^\tau \mid s^1) \Bigg] \\
&\geq \cdots \\
&\geq \sum_j \mathcal{E}_{ij}(s^0) \Bigg[\sum_{t=0}^{T} \sum_{s^t} Q_j(s^t \mid s^0) D_{ij}(\iota, s^0 \mid s^0) \\
&\qquad\qquad + Q_j(s^{T+1} \mid s^0) \sum_{\tau \geq T+1} \sum_{s^\tau} Q_j(s^\tau \mid s^{T+1}) D_{ij}(\iota, s^\tau \mid s^{T+1}) \Bigg]
\end{aligned}$$

where the second line is a rearrangement of the first, the third line uses the no-arbitrage condition $Q_j(s^t \mid s^0) = Q_j(s_t \mid s^{t-1}) \times Q_j(s_{t-1} \mid s^{t-2}) \times \cdots \times Q_j(s_1 \mid s^0) \times Q(s^0 \mid s^0)$, and the fourth line uses the flow budget constraint (47). Taking the limit as $T \to \infty$ and using the no-Ponzi scheme constraint that

$$\lim_{T \to \infty} D_{ij}(\iota, s^\tau \mid s^{T+1}) \geq 0,$$

for all $\tau > T+1$, the intertemporal budget constraint becomes

$$\bar{B}_i(\iota) \geq \sum_j \mathcal{E}_{ij}(s^0) \sum_{t=0}^{\infty} \sum_{s^t} Q_j(s^t) D_{ij}(\iota, s^t \mid s^t) = \sum_{t=0}^{\infty} \sum_{s^t} \sum_j \mathcal{E}_{ij}(s^0) Q_j(s^t) D_{ij}(\iota, s^t).$$

Rewrite this intertemporal budget constraint as

$$\sum_t \sum_{s^t} \sum_j \left[\frac{\mathcal{E}_{ij}(s^0) Q_j(s^t)}{\mathcal{E}_{ij}(s^t)} \frac{P_i^F(s^t)}{P_i^F(s^0)} \times \frac{\mathcal{E}_{ij}(s^t) D_{ij}(\iota, s^t)}{P_i^F(s^t)} \right] \leq \frac{\bar{B}_i(\iota)}{P_i^F(s^0)} \equiv \bar{b}_i(\iota).$$

Substituting the real notation in equations (51) and (52), the intertemporal budget constraint becomes

$$\sum_t \sum_{s^t} \sum_j q_{ij}(s^t) d_{ij}(\iota, s^t) \leq \bar{b}_i(\iota).$$

The cash-in-advance constraint holds with equality, so I substitute it for $C_i^g(\iota, s^t)$ in the household's utility function (1). Given the assumption that μ_i is equalized across households, it will be the case that $d_{ij}(\iota, s^t) \geq 0$, which simplifies the expression. The household maximizes

$$\sum_t \sum_{s^t} \beta^t \pi(s^t) u \left(\sum_j \frac{d_{ij}(\iota, s^t)}{Z_{ij}(\iota)} + \frac{w_i(s^t) L_i(\iota, s^t)}{Z_i^L} - v(L_i(\iota, s^t)) \right) \tag{54}$$

subject to budget constraint

$$\sum_t \sum_{s^t} \sum_j q_{ij}(s^t) d_{ij}(\iota, s^t) \leq \bar{b}_i(\iota). \tag{55}$$

Step 2: Labor supply. The household's first-order condition for $L_i(\iota, s^t)$ is

$$0 = \beta^t \pi(s^t) u_i'(\iota, s^t) \left(\frac{w_i(s^t)}{Z_i^L} - v'(L_i(\iota, s^t)) \right),$$

so

$$v'(L_i(\iota, s^t)) = \frac{w_i(s^t)}{Z_i^L}. \tag{56}$$

Note that all households ι supply the same amount of labor, as the RHS is independent of ι:

$$L_i(\iota, s^t) \equiv L_i(s^t).$$

Let the net contribution of labor to the term inside the u function be denoted by

$$u_i^L(s^t) \equiv \frac{w_i(s^t) L_i(s^t)}{Z_i^L} - v(L_i(s^t)), \tag{57}$$

where $L_i(s^t)$ satisfies the aforementioned FOC.

Step 3: Portfolio choice. In state s^t, household ι chooses its portfolio so that it receives dividend income from the country $j = J(\iota, s^t)$ that provides the highest return after paying for financial services:

$$J(\iota, s^t) = \arg\max_j \frac{1}{q_{ij}(s^t) Z_{ij}(\iota)}. \tag{58}$$

Recall that

$$\frac{1}{Z_{ij}(\iota)} \sim \text{Fréchet}\left(\frac{\kappa_0}{Z_{ij}}, \zeta - 1\right).$$

Multiplying a Fréchet random variable by a scalar returns a new Fréchet distribution:

$$\frac{1}{q_{ij}(s^t)Z_{ij}(\iota)} \sim \text{Fréchet}\left(\frac{\kappa_0}{q_{ij}(s^t)Z_{ij}}, \zeta - 1\right) \tag{59}$$

Then

$$\Pr(J(\iota, s^t) = j) = \Pr\left(j = \arg\max_{\hat{j}} \frac{1}{q_{i\hat{j}}(s^t)Z_{i\hat{j}}(\iota)}\right).$$

Using the max-stability of the Fréchet distribution, the probability becomes:

$$\Pr(J(\iota, s^t) = j) = \frac{(q_{ij}(s^t)Z_{ij})^{-(\zeta-1)}}{\sum_{\hat{j}}(q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}} \tag{60}$$

Substituting $j = J(\iota, s^t)$ into the household's utility function (54), the household's problem is to choose a portfolio $\{b_{iJ(\iota,s^t)}(\iota, s^t)$ to maximize

$$\sum_t \sum_{s^t} \beta^t \pi(s^t) u\left(\frac{d_{iJ(\iota,s^t)}(\iota, s^t)}{Z_{iJ(\iota,s^t)}(\iota)} + u_i^L(s^t)\right)$$

subject to budget constraint

$$\sum_t \sum_{s^t} q_{J(\iota,s^t)}(s^t) d_{iJ(\iota,s^t)}(\iota, s^t) \leq \bar{b}_i(\iota).$$

The household's first-order condition for $d_{ij}(\iota, s^t)$, given $j = J(\iota, s^t)$, is

$$\beta^t \pi(s^t) u'\left(\frac{d_{ij}(\iota, s^t)}{Z_{ij}(\iota)} + u_i^L(s^t)\right) \frac{1}{Z_{ij}(\iota)} = \mu_i q_{ij}(s^t)$$

where μ_i is the Lagrange multiplier on the budget constraint, and is assumed to be the same across all households in a country. Rearranging:

$$u'\left(\frac{d_{ij}(\iota, s^t)}{Z_{ij}(\iota)} + u_i^L(s^t)\right) = \left(\frac{\beta^t \pi(s^t)}{\mu_i q_{ij}(s^t) Z_{ij}(\iota)}\right)^{-1}$$

With power utility, $u'(x) = x^{-\gamma}$, this becomes

$$\frac{d_{ij}(\iota, s^t)}{Z_{ij}(\iota)} + u_i^L(s^t) = \left(\frac{\beta^t \pi(s^t)}{\mu_i q_{ij}(s^t) Z_{ij}(\iota)}\right)^{\frac{1}{\gamma}} \tag{61}$$

where $j = J(\iota, s^t)$.

Step 4: Aggregate portfolio choice. This section derives the aggregate portfolio holdings of country i. Aggregate country i holdings of country j's bonds are

$$d_{ij}(s^t) \equiv \int d_{ij}(\iota, s^t) d\iota = \mathrm{E}\big[d_{ij}(\iota, s^t) \mid J(\iota) = j\big] \times \Pr\big(J(\iota) = j\big), \tag{62}$$

where the last expression makes use of the law of iterated expectation. First, consider $\mathrm{E}[d_{ij}(\iota, s^t) \mid J(\iota) = j]$, which is the average amount of j's claims held by households who choose to hold j's claims. Multiplying equation (61) through by $Z_{ij}(\iota)$ and taking the conditional expectation:

$$\begin{aligned} &\mathrm{E}\big[d_{ij}(\iota, s^t) \mid J(\iota) = j\big] + u_i^L(s^t)\, \mathrm{E}\big[Z_{ij}(\iota) \mid J(\iota) = j\big] \\ &= \left(\frac{\beta^t \pi(s^t)}{\mu_i q_{ij}(s^t)}\right)^{\frac{1}{\gamma}} \mathrm{E}\big[Z_{ij}(\iota)^{1-\frac{1}{\gamma}} \mid J(\iota) = j\big] \end{aligned} \tag{63}$$

To evaluate these conditional expectations, I need the obtain the conditional distributions. The max-stability of the Fréchet distribution, combined with the decision rule for $J(\iota, s^t)$ in (58) and the distribution of $1/q_{ij}(s^t) Z_{ij}(\iota)$ in (59), obtains the following conditional distribution:

$$\frac{1}{q_{ij}(s^t) Z_{ij}(\iota)} \;\bigg|\; J(\iota, s^t) = j \sim \text{Fréchet}\left(\kappa_0 \left(\sum_{\hat{j}} (q_{i\hat{j}}(s^t) Z_{i\hat{j}})^{-(\zeta-1)}\right)^{\frac{1}{\zeta-1}},\ \zeta - 1\right).$$

Multiplying a Fréchet random variable by the scalar $q_{ij}(s^t)$ returns a new Fréchet distribution:

$$\frac{1}{Z_{ij}(\iota)} \;\bigg|\; J(\iota, s^t) = j \sim \text{Fréchet}\left(\kappa_0 q_{ij} \left(\sum_{\hat{j}} (q_{i\hat{j}}(s^t) Z_{i\hat{j}})^{-(\zeta-1)}\right)^{\frac{1}{\zeta-1}},\ \zeta - 1\right) \tag{64}$$

Taking the reciprocal of a Fréchet random variable obtains a Weibull random variable:

$$Z_{ij}(\iota) \mid J(\iota, s^t) = j \sim \text{Weibull}\left(\frac{1}{\kappa_0 q_{ij}} \left(\sum_{\hat{j}} (q_{i\hat{j}}(s^t) Z_{i\hat{j}})^{-(\zeta-1)}\right)^{-\frac{1}{\zeta-1}},\ \zeta - 1\right)$$

$$Z_{ij}(\iota)^{1-\frac{1}{\gamma}} \mid J(\iota, s^t) = j \sim \text{Weibull}\left(\frac{1}{(\kappa_0 q_{ij})^{1-\frac{1}{\gamma}}} \left(\sum_{\hat{j}} (q_{i\hat{j}}(s^t) Z_{i\hat{j}})^{-(\zeta-1)}\right)^{-\frac{1}{\zeta-1}(1-\frac{1}{\gamma})},\ \frac{\zeta - 1}{1 - \frac{1}{\gamma}}\right)$$

Evaluating the expectations:

$$\mathrm{E}\big[Z_{ij}(\iota) \mid J(\iota, s^t) = j\big] = \frac{1}{\kappa_0 q_{ij}} \left(\sum_{\hat{j}} (q_{i\hat{j}}(s^t) Z_{i\hat{j}})^{-(\zeta-1)}\right)^{-\frac{1}{\zeta-1}} \Gamma(1 + \tfrac{1}{\zeta-1})$$

$$\mathrm{E}\big[Z_{ij}(\iota)^{1-\frac{1}{\gamma}} \mid J(\iota, s^t) = j\big] = \frac{1}{(\kappa_0 q_{ij})^{1-\frac{1}{\gamma}}} \left(\sum_{\hat{j}} (q_{i\hat{j}}(s^t) Z_{i\hat{j}})^{-(\zeta-1)}\right)^{-\frac{1}{\zeta-1}(1-\frac{1}{\gamma})} \Gamma\big(1 + \tfrac{1}{\zeta-1}(1 - \tfrac{1}{\gamma})\big)$$

Simplify: $\Gamma(1+\frac{1}{\zeta-1}) = \Gamma(\frac{\zeta}{\zeta-1})$ and $\Gamma(1+\frac{1}{\zeta-1}(1-\frac{1}{\gamma})) = \Gamma(\frac{\zeta-1/\gamma}{\zeta-1})$. Now substitute these expressions back into equation (63):

$$\begin{aligned} &\mathrm{E}[d_{ij}(\iota, s^t) \mid J(\iota) = j] + \frac{u_i^L(s^t)}{q_{ij}(s^t)}\left(\sum_{\hat{j}}(q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}\right)^{-\frac{1}{\zeta-1}} \times \frac{\Gamma(\frac{\zeta}{\zeta-1})}{\kappa_0} \\ &= \left(\frac{\beta^t\pi(s^t)}{\mu_i}\right)^{\frac{1}{\gamma}} \frac{1}{q_{ij}(s^t)}\left(\sum_{\hat{j}}(q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}\right)^{-\frac{1}{\zeta-1}(1-\frac{1}{\gamma})} \times \frac{\Gamma(\frac{\zeta-1/\gamma}{\zeta-1})}{\kappa_0^{1-\frac{1}{\gamma}}} \end{aligned}$$

Multiplying this expression by $\Pr(J(\iota) = j)$, using the expression derived in equation (60), and using the expression for aggregate holdings of j's claims in equation (62) to simplify the first term on the left-hand side:

$$\begin{aligned} &d_{ij}(s^t) + \frac{u_i^L(s^t)}{q_{ij}(s^t)^\zeta Z_{ij}^{\zeta-1}}\left(\sum_{\hat{j}}(q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}\right)^{-\frac{\zeta}{\zeta-1}} \times \frac{\Gamma(\frac{\zeta}{\zeta-1})}{\kappa_0} \\ &= \left(\frac{\beta^t\pi(s^t)}{\mu_i}\right)^{\frac{1}{\gamma}} \frac{1}{q_{ij}(s^t)^\zeta Z_{ij}^{\zeta-1}}\left(\sum_{\hat{j}}(q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}\right)^{-\frac{1}{\zeta-1}(1-\frac{1}{\gamma})-1} \times \frac{\Gamma(\frac{\zeta-1/\gamma}{\zeta-1})}{\kappa_0^{1-\frac{1}{\gamma}}} \end{aligned}$$

Separate out the terms involving j from the terms that do not involve j:

$$\begin{aligned} d_{ij}(s^t) &= \frac{1}{q_{ij}(s^t)^\zeta Z_{ij}^{\zeta-1}}\left(\sum_{\hat{j}}(q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}\right)^{-\frac{\zeta}{\zeta-1}} \times \\ &\quad \left[\left(\frac{\beta^t\pi(s^t)}{\mu_i}\right)^{\frac{1}{\gamma}}\left(\sum_{\hat{j}}(q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}\right)^{\frac{1}{\zeta-1}\frac{1}{\gamma}} \times \frac{\Gamma(\frac{\zeta-1/\gamma}{\zeta-1})}{\kappa_0^{1-\frac{1}{\gamma}}} - u_i^L(s^t) \times \frac{\Gamma(\frac{\zeta}{\zeta-1})}{\kappa_0}\right] \\ &= \frac{1}{q_{ij}(s^t)^\zeta Z_{ij}^{\zeta-1}}\left(\sum_{\hat{j}}(q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}\right)^{-\frac{\zeta}{\zeta-1}} A_{0i}(s^t) \end{aligned}$$

where I denote the bracketed term on the second line by $A_{0i}(s^t)$:

$$A_{0i}(s^t) = \left(\frac{\beta^t\pi(s^t)}{\mu_i}\right)^{\frac{1}{\gamma}}\left(\sum_{\hat{j}}(q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}\right)^{\frac{1}{\zeta-1}\frac{1}{\gamma}} \times \frac{\Gamma(\frac{\zeta-1/\gamma}{\zeta-1})}{\kappa_0^{1-\frac{1}{\gamma}}} - u_i^L(s^t) \times \frac{\Gamma(\frac{\zeta}{\zeta-1})}{\kappa_0}. \quad (65)$$

Divide by Z_{ij}:

$$\frac{d_{ij}(s^t)}{Z_{ij}} = (q_{ij}(s^t)Z_{ij})^{-\zeta}\left(\sum_{\hat{j}}(q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}\right)^{-\frac{\zeta}{\zeta-1}} A_{0i}(s^t). \quad (66)$$

Take the power of $(\zeta-1)/\zeta$:

$$\left(\frac{d_{ij}(s^t)}{Z_{ij}}\right)^{\frac{\zeta-1}{\zeta}} = (q_{ij}(s^t)Z_{ij})^{-(\zeta-1)}\left(\sum_{\hat{j}}(q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}\right)^{-1} A_{0i}(s^t)^{\frac{\zeta-1}{\zeta}}.$$

Sum over j:

$$\sum_j \left(\frac{d_{ij}(s^t)}{Z_{ij}}\right)^{\frac{\zeta-1}{\zeta}} = \sum_j (q_{ij}(s^t)Z_{ij})^{-(\zeta-1)}\left(\sum_{\hat{j}}(q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}\right)^{-1} A_{0i}(s^t)^{\frac{\zeta-1}{\zeta}} = A_{0i}(s^t)^{\frac{\zeta-1}{\zeta}}.$$

Define the bond aggregator d_i^{agg} as the LHS of the preceding expression taken to the power of $\zeta/(\zeta-1)$:

$$d_i^{agg}(s^t) \equiv \left[\sum_j \left(\frac{d_{ij}(s^t)}{Z_{ij}}\right)^{\frac{\zeta-1}{\zeta}}\right]^{\frac{\zeta}{\zeta-1}} = A_{0i}(s^t). \tag{67}$$

Dividing equation (66) by (67) obtains:

$$\frac{d_{ij}(s^t)/Z_{ij}}{d_i^{agg}(s^t)} = (q_{ij}(s^t)Z_{ij})^{-\zeta}\left(\sum_{\hat{j}}(q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}\right)^{-\frac{\zeta}{\zeta-1}}$$

Hence, aggregate holdings of j's bonds by i's households are most easily represented as shares of the aggregator d_i^{agg}. The previous expression rearranges to:

$$\frac{d_{ij}(s^t)/Z_{ij}}{d_i^{agg}(s^t)} = \left(\frac{(q_{ij}(s^t)Z_{ij})^{-(\zeta-1)}}{\sum_{\hat{j}}(q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}}\right)^{\frac{\zeta}{\zeta-1}}. \tag{68}$$

Now substituting the relationship between q_{ij} and R_{ij}^D in equation (52), this becomes

$$\begin{aligned} \frac{d_{ij}(s^t)/Z_{ij}}{d_i^{agg}(s^t)} &= \left(\frac{(R_{ij}^D(s^t)P_i^F(s^t)/Z_{ij}P_i^F(s^0))^{\zeta-1}}{\sum_{\hat{j}}(R_{i\hat{j}}^D(s^t)P_i^F(s^t)/Z_{i\hat{j}}P_i^F(s^0))^{\zeta-1}}\right)^{\frac{\zeta}{\zeta-1}} \\ &= \frac{(R_{ij}^D(s^t)/Z_{ij})^{\zeta}}{\left(\sum_{\hat{j}}(R_{i\hat{j}}^D(s^t)/Z_{i\hat{j}})^{\zeta-1}\right)^{\frac{\zeta}{\zeta-1}}} \end{aligned} \tag{69}$$

Country i's aggregate portfolio return satisfies

$$\frac{1}{R_i^H(s^t)} = \sum_j \frac{d_{ij}(s^t)/Z_{ij}}{d_i^{agg}(s^t)}\frac{1}{R_{ij}^D/Z_{ij}} = \sum_j \frac{(R_{ij}^D(s^t)/Z_{ij})^{\zeta-1}}{\left(\sum_{\hat{j}}(R_{i\hat{j}}^D(s^t)/Z_{i\hat{j}})^{\zeta-1}\right)^{\frac{\zeta}{\zeta-1}}}$$

which simplifies to

$$\frac{1}{R_i^H(s^t)} = \left(\sum_j (R_{ij}^D(s^t)/Z_{ij})^{\zeta-1}\right)^{1-\frac{\zeta}{\zeta-1}} = \left(\sum_j (R_{ij}^D(s^t)/Z_{ij})^{\zeta-1}\right)^{-\frac{1}{\zeta-1}}$$

so

$$R_i^H(s^t) = \left(\sum_j (R_{ij}^D(s^t)/Z_{ij})^{\zeta-1}\right)^{\frac{1}{\zeta-1}}$$

obtains the desired result in equation (50). Substituting this into the denominator of the portfolio dividend share (69) obtains

$$\frac{d_{ij}(s^t)/Z_{ij}}{d_i^{agg}(s^t)} = \left(\frac{R_{ij}^D(s^t)/Z_{ij}}{R_i^H(s^t)}\right)^{\zeta},$$

which obtains the desired result in equation (49). *QED*

A.2 Proof of Proposition 2 (Aggregation)

Restatement of proposition: given prices, country i's aggregate consumption $C_i(s^t)$, labor $L_i(s^t)$, portfolios $B_{ij}(s^t)$, and dividend income $D_{ij}(s^t)$ are identical to those of the following representative household economy: the representative household chooses consumption of goods $C_i^g(s^t)$, labor $L_i(s^t)$, and portfolio $B_{ij}(s^t)$ to maximize

$$\sum_t \sum_{s^t} \beta^t \pi(s^t) u\big(C_i^g(s^t) - v(L_i(s^t))\big), \tag{70}$$

subject to cash-in-advance constraint

$$P_i^F(s^t)C_i^g(s^t) = \kappa_1 D_i^{agg}(s^t) + \frac{W_i(s^t)L_i(s^t)}{Z_i^L}, \tag{71}$$

where $\kappa_1 = \kappa_0/\Gamma(\frac{\zeta}{\zeta-1}) \approx 1$ and the dividend income aggregator D_i^{agg} is defined in equation (27)*, subject to a sequence of budget constraints in each country j's financial market*

$$\sum_{\tau\geq t}\sum_{s^\tau} Q_j(s^\tau \mid s^t)D_{ij}(s^\tau \mid s^t) \leq \sum_{\tau\geq t}\sum_{s^\tau} Q_j(s^\tau \mid s^t)D_{ij}(s^\tau \mid s^{t-1}), \tag{72}$$

time-0 budget constraint

$$\sum_t \sum_{s^t} \sum_j \mathcal{E}_{ij}(s^0)Q_j(s^t)D_{ij}(\iota, s^t \mid s^0) \leq \bar{B}_i(\iota) \tag{73}$$

where the initial endowment is $\bar{B}_i = \int \bar{B}_i(\iota)d\iota$, and the no-Ponzi scheme constraint $\lim_{\tau\to\infty} D_{ij}(s^\tau \mid s^t) \geq 0$ in each j.

Step 1: Representative household's problem. As with the heterogeneous households, it is convenient to reformulate the problem in real terms, using the variables defined in (51)–(53), and

$$d_i^{agg}(s^t) \equiv \frac{D_i^{agg}(s^t)}{P_i^F(s^t)} = \left[\sum_j \left(\frac{d_{ij}(s^t)}{Z_{ij}}\right)^{\frac{\zeta-1}{\zeta}}\right]^{\frac{\zeta}{\zeta-1}}. \tag{74}$$

Using this notation, the household's cash-in-advance constraint (71) becomes

$$C_i^g(s^t) \leq \kappa_1 d_i^{agg}(s^t) + \frac{W_i(s^t)L_i(s^t)}{Z_i^L}.$$

I assume that monetary settings are such that this constraint holds with equality. Substituting this expression for $C_i^g(s^t)$ into the household's utility function (70), the household maximizes

$$\sum_t \sum_{s^t} \beta^t \pi(s^t) u\left(\kappa_1 d_i^{agg}(s^t) + \frac{w_i(s^t) L_i(s^t)}{Z_i^L} - v(L_i(s^t)) \right) \tag{75}$$

subject to budget constraint

$$\sum_t \sum_{s^t} \sum_j q_{ij}(s^t) d_{ij}(s^t) \leq \bar{b}_i.$$

The sequential budget constraints (72) and (73) are collected into an intertemporal budget constraint in the same manner as Step 1 in Appendix A.1.

Step 2: Representative household's labor supply. The representative household's first-order condition for $L_i(s^t)$ is

$$0 = \beta^t \pi(s^t) u_i'(s^t) \left(\frac{w_i(s^t)}{Z_i^L} - v'(L_i(s^t)) \right)$$

so the FOC is the same as the heterogeneous agent problem in equation (56):

$$v'(L_i(s^t)) = \frac{w_i(s^t)}{Z_i^L}. \tag{76}$$

As with the heterogeneous household case, let the net contribution of labor to the term inside the u function be denoted by

$$u_i^L(s^t) \equiv \frac{w_i(s^t) L_i(s^t)}{Z_i^L} - v(L_i(s^t)).$$

Step 3: Representative household's bond holdings. The representative household's first-order condition for $d_{ij}(s^t)$ is

$$\beta^t \pi(s^t) u'\big(\kappa_1 d_i^{agg}(s^t) + u_i^L(s^t) \big) \left(\frac{d_i^{agg}(s^t)}{d_{ij}(s^t)/Z_{ij}} \right)^{-\frac{1}{\zeta}} \frac{1}{Z_{ij}} = \mu_i q_{ij}(s^t) \tag{77}$$

where μ_i is the Lagrange multiplier on the budget constraint. Using the power utility form $u'(x) = x^{-\gamma}$, and rearranging:

$$\left(\frac{d_{ij}(s^t)/Z_{ij}}{d_i^{agg}(s^t)} \right)^{\frac{1}{\zeta}} = \frac{\beta^t \pi(s^t)}{\mu_i} \frac{1}{q_{ij}(s^t) Z_{ij}} \big(\kappa_1 d_i^{agg}(s^t) + u_i^L(s^t) \big)^{-\gamma} \tag{78}$$

so

$$\left(\frac{d_{ij}(s^t)}{Z_{ij}} \right)^{\frac{1}{\zeta}} = \frac{\beta^t \pi(s^t)}{\mu_i} \frac{1}{q_{ij}(s^t) Z_{ij}} \big(\kappa_1 d_i^{agg}(s^t) + u_i^L(s^t) \big)^{-\gamma} d_i^{agg}(s^t)^{\frac{1}{\zeta}}$$

Taking the power of $\zeta - 1$:

$$\left(\frac{d_{ij}(s^t)}{Z_{ij}}\right)^{\frac{\zeta-1}{\zeta}} = \left(\frac{\beta^t \pi(s^t)}{\mu_i}\right)^{\zeta-1} (q_{ij}(s^t)Z_{ij})^{-(\zeta-1)} \left(\kappa_1 d_i^{agg}(s^t) + u_i^L(s^t)\right)^{-\gamma(\zeta-1)} d_i^{agg}(s^t)^{\frac{\zeta-1}{\zeta}}$$

Sum over j:

$$\sum_j \left(\frac{d_{ij}(s^t)}{Z_{ij}}\right)^{\frac{\zeta-1}{\zeta}} = \left(\frac{\beta^t \pi(s^t)}{\mu_i}\right)^{\zeta-1} \left(\sum_j (q_{ij}(s^t)Z_{ij})^{-(\zeta-1)}\right) (\kappa_1 d_i^{agg}(s^t) + u_i^L(s^t))^{-\gamma(\zeta-1)} d_i^{agg}(s^t)^{\frac{\zeta-1}{\zeta}}$$

Take the power of $\zeta/(\zeta-1)$ and use the definition of d_i^{agg} in equation (74):

$$d_i^{agg}(s^t) = \left(\frac{\beta^t \pi(s^t)}{\mu_i}\right)^{\zeta} \left(\sum_j (q_{ij}(s^t)Z_{ij})^{-(\zeta-1)}\right)^{\frac{\zeta}{\zeta-1}} \left(\kappa_1 d_i^{agg}(s^t) + u_i^L(s^t)\right)^{-\gamma\zeta} d_i^{agg}(s^t)$$

The terms $d_i^{agg}(s^t)$ cancel out on the LHS and RHS. Rearranging and taking the power of $1/\zeta$:

$$\left(\kappa_1 d_i^{agg}(s^t) + u_i^L(s^t)\right)^{-\gamma} = \left(\frac{\beta^t \pi(s^t)}{\mu_i}\right)^{-1} \left(\sum_j (q_{ij}(s^t)Z_{ij})^{-(\zeta-1)}\right)^{-\frac{1}{\zeta-1}}. \tag{79}$$

Substituting this into equation (78):

$$\left(\frac{d_{ij}(s^t)/Z_{ij}}{d_i^{agg}(s^t)}\right)^{\frac{1}{\zeta}} = \frac{1}{q_{ij}(s^t)Z_{ij}} \left(\sum_j (q_{ij}(s^t)Z_{ij})^{-(\zeta-1)}\right)^{-\frac{1}{\zeta-1}}. \tag{80}$$

Taking the power of ζ, the following expression shows that the share of capital income the representative household in i receives from country j:

$$\frac{d_{ij}(s^t)/Z_{ij}}{d_i^{agg}(s^t)} = \left(\frac{(q_{ij}(s^t)Z_{ij})^{-(\zeta-1)}}{\sum_j (q_{ij}(s^t)Z_{ij})^{-(\zeta-1)}}\right)^{\frac{\zeta}{\zeta-1}} \tag{81}$$

is identical to equation (68) for the aggregated heterogeneous households.

Take the expression for the household's marginal utility in equation (79) to the power of $-1/\gamma$:

$$\kappa_1 d_i^{agg}(s^t) + u_i^L(s^t) = \left(\frac{\mu_i}{\beta^t \pi(s^t)}\right)^{-\frac{1}{\gamma}} \left(\sum_j (q_{ij}(s^t)Z_{ij})^{-(\zeta-1)}\right)^{\frac{1}{\zeta-1}\frac{1}{\gamma}}$$

Therefore,

$$d_i^{agg}(s^t) = \left(\frac{\beta^t \pi(s^t)}{\mu_i}\right)^{\frac{1}{\gamma}} \left(\sum_j (q_{ij}(s^t)Z_{ij})^{-(\zeta-1)}\right)^{\frac{1}{\zeta-1}\frac{1}{\gamma}} \frac{1}{\kappa_1} - \frac{u_i^L(s^t)}{\kappa_1} \tag{82}$$

Recall that in the heterogeneous households case, by combining equations (67) and (65), the bond aggregator was

$$d_i^{agg}(s^t) = \left(\frac{\beta^t \pi(s^t)}{\mu_i}\right)^{\frac{1}{\gamma}} \left(\sum_{\hat{j}} (q_{i\hat{j}}(s^t)Z_{i\hat{j}})^{-(\zeta-1)}\right)^{\frac{1}{\zeta-1}\frac{1}{\gamma}} \times \frac{\Gamma(\frac{\zeta-1/\gamma}{\zeta-1})}{\kappa_0^{1-\frac{1}{\gamma}}} - u_i^L(s^t) \times \frac{\Gamma(\frac{\zeta}{\zeta-1})}{\kappa_0}. \tag{83}$$

Step 4: Solve for the constants. By matching coefficients, we can get the expressions for $d_i^{agg}(s^t)$ to align exactly between the two cases:

$$\frac{\Gamma(\frac{\zeta-1/\gamma}{\zeta-1})}{\kappa_0^{1-\frac{1}{\gamma}}} = \frac{1}{\kappa_1}$$
$$\frac{\Gamma(\frac{\zeta}{\zeta-1})}{\kappa_0} = \frac{1}{\kappa_1}$$

Combining the two equations obtains

$$\frac{\Gamma(\frac{\zeta-1/\gamma}{\zeta-1})}{\kappa_0^{1-\frac{1}{\gamma}}} = \frac{\Gamma(\frac{\zeta}{\zeta-1})}{\kappa_0}$$

Cancelling terms and rearranging:

$$\kappa_0^{\frac{1}{\gamma}} = \frac{\Gamma(\frac{\zeta}{\zeta-1})}{\Gamma(\frac{\zeta-1/\gamma}{\zeta-1})}$$

so

$$\kappa_0 = \left(\frac{\Gamma(\frac{\zeta}{\zeta-1})}{\Gamma(\frac{\zeta-1/\gamma}{\zeta-1})}\right)^{\gamma} \tag{84}$$

and

$$\kappa_1 = \frac{\kappa_0}{\Gamma(\frac{\zeta}{\zeta-1})} = \frac{\left[\Gamma(\frac{\zeta}{\zeta-1})\right]^{\gamma-1}}{\left[\Gamma(\frac{\zeta-1/\gamma}{\zeta-1})\right]^{\gamma}} \tag{85}$$

Comparing equations (56) and (76) shows that labor supply $L_i(s^t)$ is the same across the heterogeneous and representative household economies. Comparing equations (82) and (83), along with (84) and (85), show that the aggregator $d_i^{agg}(s^t)$ is the same up to the marginal utility of wealth term μ_i. Comparing equations (68) and (81) shows that portfolio dividend shares $(d_{ij}(s^t)/Z_{ij})/d_i^{agg}(s^t)$ are the same. Given that the representative agents have the same wealth as the aggregate of the heterogeneous agents ($\bar{B}_i = \int \bar{B}_i(\iota)d\iota$), for the time-0 budget constraints to hold, it must be that the portfolios are the same, so μ_i and $d_i^{agg}(s^t)$ are the same. *QED*

A.3 Duration

In steady state, the household's portfolio pays out $\bar{D}_{ij}$ units of dividends from country j in each period. At time 0, the present value of the time-t dividend payment is $Q(s^t)\bar{D}_{ij} = \beta^t Q(s^0)\bar{D}_{ij}$. Hence, the duration of these payoffs are given by

$$\frac{\sum_{t\geq 1} t\beta^t Q(s^0)\bar{D}_{ij}}{\sum_{t\geq 1} \beta^t Q(s^0)\bar{D}_{ij}} = \frac{\sum_t t\beta^t}{\sum_t \beta^t} = \frac{\frac{1}{(1-\beta)^2}}{\frac{1}{1-\beta}} = \frac{1}{1-\beta}.$$

A.4 Shipping Costs

Say if the export of intermediate goods involves an iceberg cost, so that if one unit of j's good is shipped to country i, country i's final goods producer receives $1/Z_{ij}^X < 1$ units of the good. Fixing the state s^t, the final goods producer's problem becomes to maximize

$$P_i^F\left[\sum_j (\eta_{ij})^{\frac{1}{\theta}}\left(\frac{X_{ij}}{Z_{ij}^X}\right)^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}} - \sum_j \mathcal{E}_{ij} P_j^X X_{ij}$$

Now define $\hat{\eta}_{ij}$ as follows:

$$(\hat{\eta}_{ij})^{\frac{1}{\theta}} = (\eta_{ij})^{\frac{1}{\theta}}\left(\frac{1}{Z_{ij}^X}\right)^{\frac{\theta-1}{\theta}} \implies \hat{\eta}_{ij} = \frac{\eta_{ij}}{(Z_{ij}^X)^{\theta-1}} \tag{86}$$

Then the final goods producer's problem becomes to maximize

$$P_i^F\left[\sum_j (\hat{\eta}_{ij})^{\frac{1}{\theta}}(X_{ij})^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}} - \sum_j \mathcal{E}_{ij} P_j^X X_{ij}.$$

This has the same form as the final goods producer's problem without trade costs (see (17)), with η_{ij} replaced by a composite parameter of the pure input weights η_{ij} and the iceberg trade cost.

B Variants of the Model

B.1 Financial Frictions: Limiting Cases

The case with no financial frictions is recovered by setting $\zeta = \infty$ and $Z_{ij} = 1$ for all i and j. The case with financial autarky is recovered by setting $Z_{ij} = \infty$ for all $j \neq i$.

B.2 Differentiated Capital Goods

Oviedo and Singh (2013) provide evidence that capital goods have a higher import composition than final consumption goods. This section extends the model to capture their finding. For final consumption goods, the final goods producer has the production function

$$C_i(s^t) = \left[\sum_j \eta_{ij}^F(s^t)^{\frac{1}{\theta}} X_{ij}^F(s^t)^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}}, \tag{87}$$

where $\eta_{ij}^F(s^t)$ is the weight of j's intermediate good in i's final good and θ is the elasticity of substitution between each country's intermediate good, or the *trade elasticity* for short. The corresponding production function for capital goods is

$$I_i^K(s^t) = \left[\sum_j \eta_{ij}^K(s^t)^{\frac{1}{\theta}} X_{ij}^K(s^t)^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}}, \tag{88}$$

with $\eta_{ij}^K > \eta_{ij}^F$ for $j \neq i$.

Relative demand shocks ω_{jt} shift the weight of country j's intermediate good in producing each country i's final good and capital good:

$$\eta^F_{ij}(s^t) = \frac{\bar{\eta}^F_{ij}\exp(\omega_{jt})}{\sum_{\hat{j}}\bar{\eta}^F_{i\hat{j}}\exp(\omega_{\hat{j}t})}, \qquad \eta^K_{ij}(s^t) = \frac{\bar{\eta}^K_{ij}\exp(\omega_{jt})}{\sum_{\hat{j}}\bar{\eta}^K_{i\hat{j}}\exp(\omega_{\hat{j}t})}.$$

The final goods producer's problem: in the final consumption goods market, it chooses intermediates demand $X^F_{ij}(s^t)$ to maximize profits

$$P^F_i(s^t)\left[\sum_j \eta^F_{ij}(s^t)^{\frac{1}{\theta}} X^F_{ij}(s^t)^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}} - \sum_j \mathcal{E}_{ij}(s^t)P^X_j(s^t)X^F_{ij}(s^t) \tag{89}$$

As in the baseline model, intermediate goods follows the law of one price: in each country i, j's good is priced at $\mathcal{E}_{ij}(s^t)P^X_j(s^t)$ in units of i's currency. The problem in the capital goods market is similar.

The final goods production function (87) describes the market clearing condition for each country i's final consumption goods

$$C_i(s^t) = \left[\sum_j \eta^F_{ij}(s^t)^{\frac{1}{\theta}} X^F_{ij}(s^t)^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}}. \tag{90}$$

The capital good production function (88) describes market clearing for capital goods:

$$I^K_i(s^t) = \left[\sum_j \eta^K_{ij}(s^t)^{\frac{1}{\theta}} X^K_{ij}(s^t)^{\frac{\theta-1}{\theta}}\right]^{\frac{\theta}{\theta-1}}. \tag{91}$$

Market clearing for intermediate goods made by country j is now

$$\sum_{i=1}^N \left[X^F_{ij}(s^t) + X^K_{ij}(s^t)\right] = A_j(s^t)K_j(s^t)^\alpha L_j(s^t)^{1-\alpha} \quad \text{for all } j = 1, \dots, N. \tag{92}$$

C Data Appendix

C.1 Data Construction

International data come from International Financial Statistics (IMF, 2025), except for market capitalization data used to calculate the denominator of the home bias measure in equation (44), which are compiled by World Federation of Exchanges (2025) and published by the World Bank. US consumption of financial services is from the Bureau of Economic Analysis (2025), Table 2.4.5. Pre-Euro currency exchange rates are from Chari, Kehoe and McGrattan (2002). For trade expenditure, I use data from the IMF's Direction of Trade Statistics.

C.2 Plots

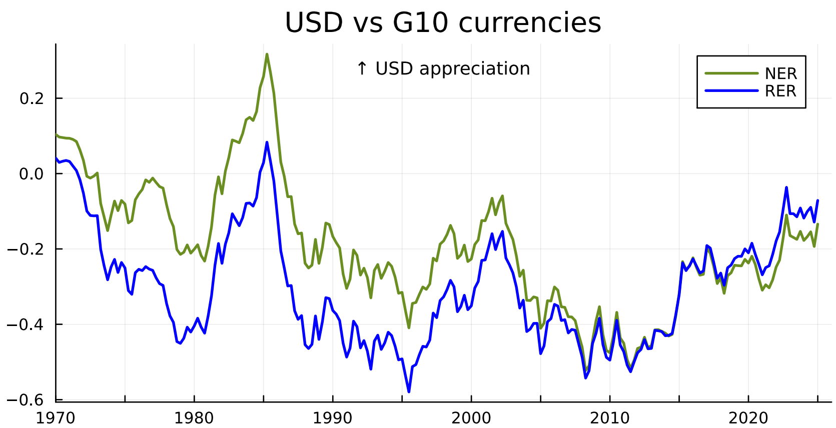

Figure 12: US Dollar Nominal and Real Exchange Rates vs G10 Currencies

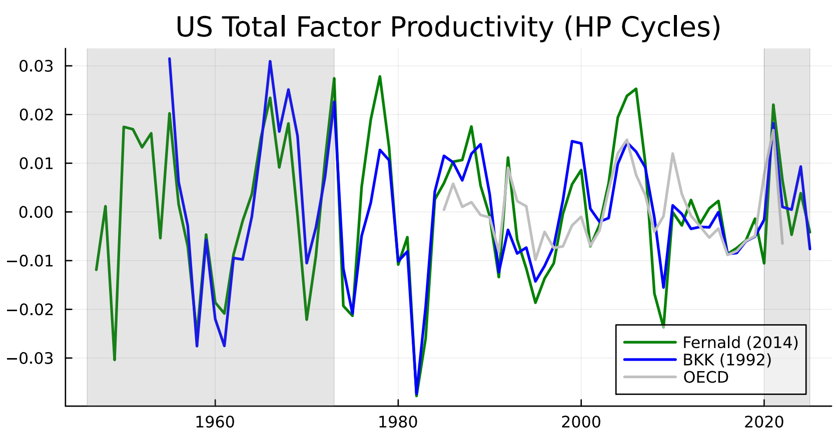

Figure 13: US Total Factor Productivity: Comparison of Sources

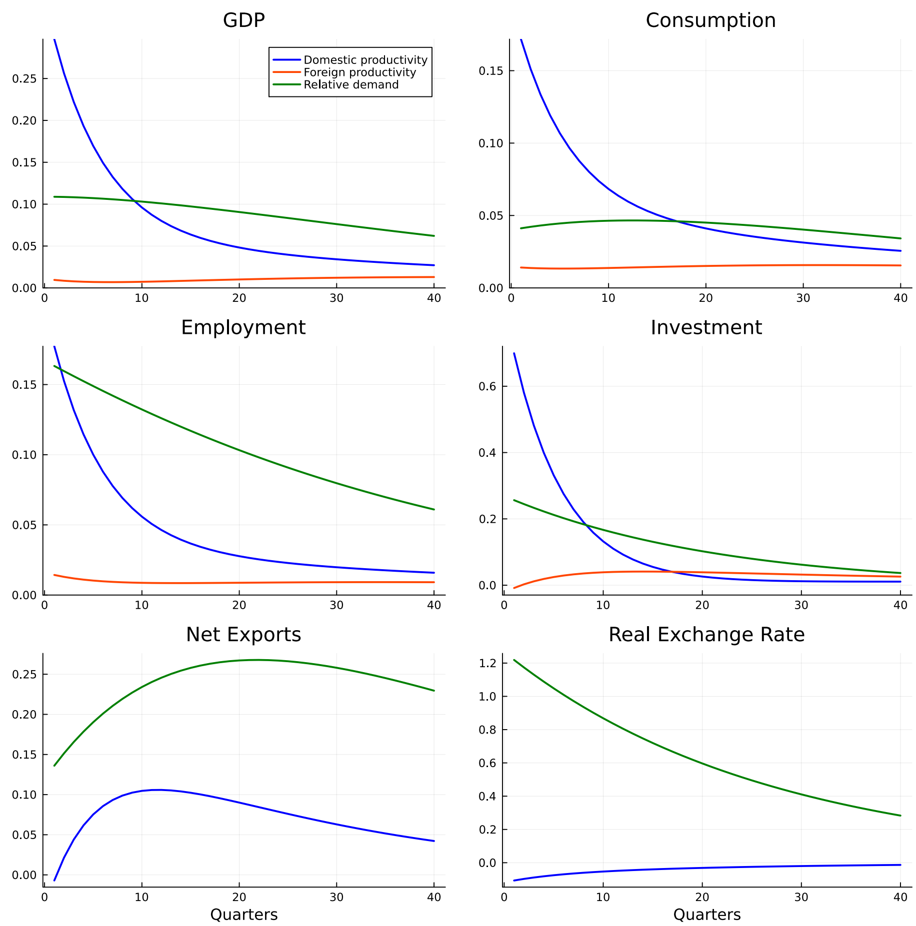

Figure 14: Impulse Response Functions

Notes: Percentage deviations from steady state in response to 1 s.d. shocks to domestic productivity (blue), foreign productivity (orange), and relative demand (green).

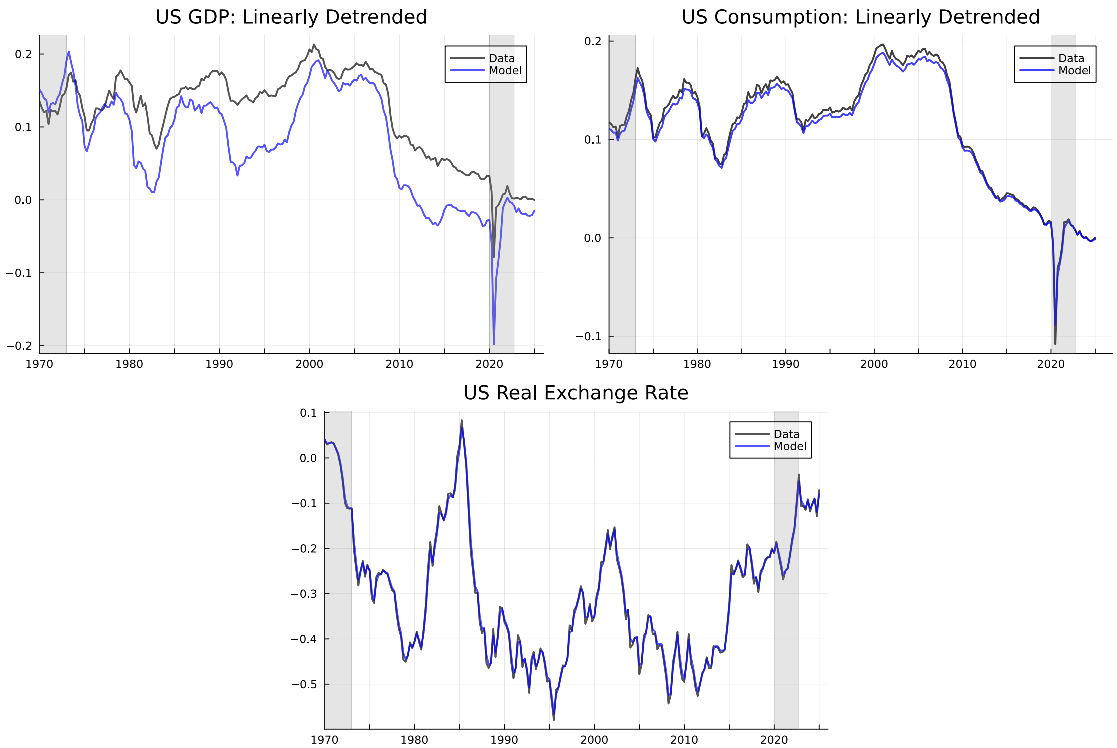

Figure 15: Kalman Filter: Model vs Data

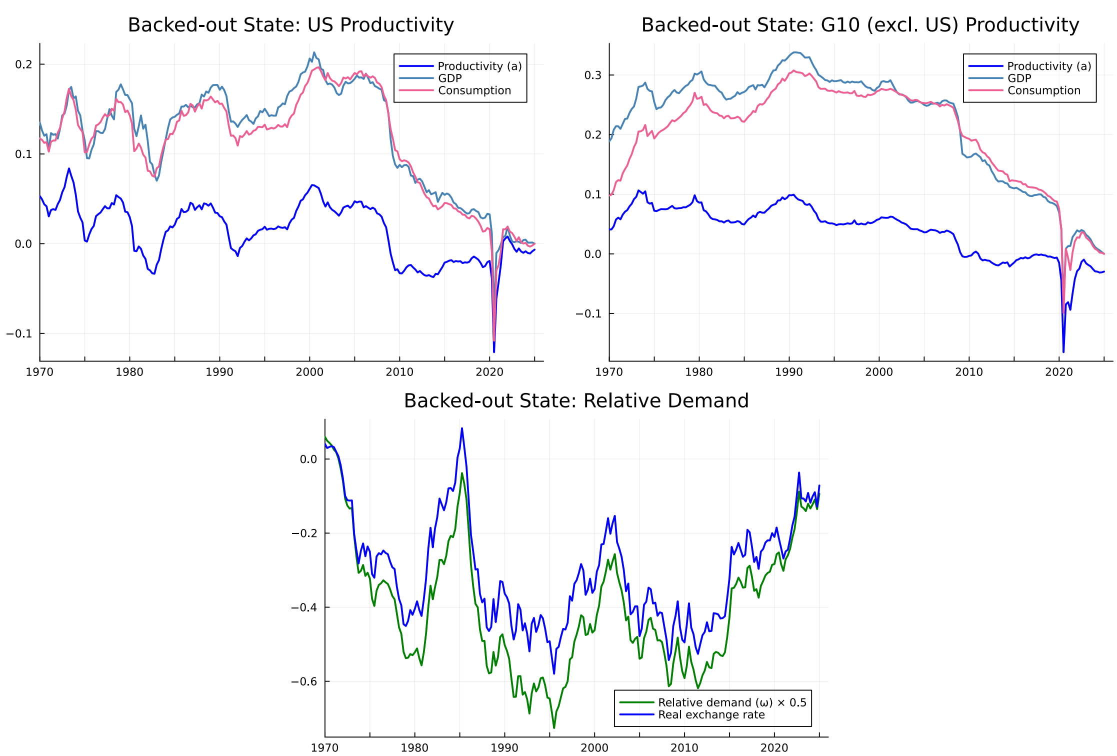

Figure 16: Kalman Filter: Time Series of Shocks